Workshop

PyPSA-Eur: A Sector-Coupled Open Optimisation Model of the European Energy System

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INDEPENDENT RESEARCH FUND DENMARK

github.com/pypsa/pypsa-eur
pypsa-eur.readthedocs.io

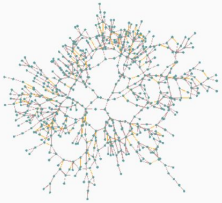
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What is PyPSA?

Our research focus:

- Cost-effective pathways to reduce greenhouse gas emissions
- Evaluation of grid expansion, hydrogen strategies, carbon management strategies
- Co-optimisation of generation, storage, conversion and transmission infrastructure
- Algorithms to improve the tractability of models
- All open source and open data

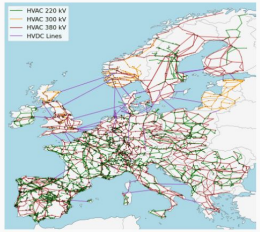
PyPSA



A python software toolbox for simulating and optimising modern power systems.

Documentation »

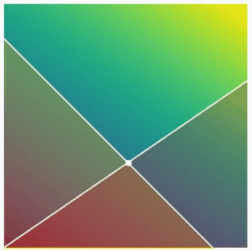
PyPSA-Eur



A Sector-Coupled Open Optimisation Model of the European Energy System

Documentation »

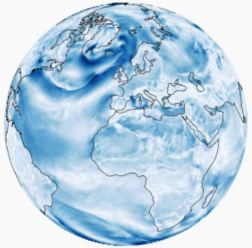
Linopy



Linear optimization interface for N-D labeled variables.

Documentation »

Atlite



A Lightweight Python Package for Calculating Renewable Power Potentials and Time Series

Documentation »

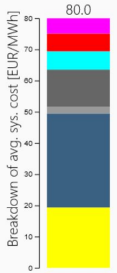
Powerplantmatching



A toolset for cleaning, standardizing and combining multiple power plant databases.

Documentation »

Model Energy



An online toolkit for calculating renewable electricity supplies.

Application examples

TSOs





NGOs and international organisations







Regulators

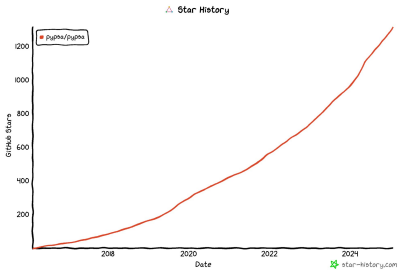


International



https://docs.pypsa.org/latest/home/users/

PyPSA:
Python for Power System Analysis



Capabilities

Capacity expansion (linear)
- single-horizon
- multi-horizon

Market modelling (linear)
- Linear optimal power flow
- Security-constrained LOPF
- Unit commitment
- Dispatch & redispatch

Non-linear power flow
Modelling-to-Generate-Alternatives
Stochastic Optimisation

With components for

- Electricity transmission networks and pipelines.
- Generators with unit commitment constraints
- Variable generation with time series (e.g. wind and solar)
- Storage with efficiency losses and inflow/spillage for hydro
- Conversion between energy carriers (PtX, CHP, BEV, DAC)

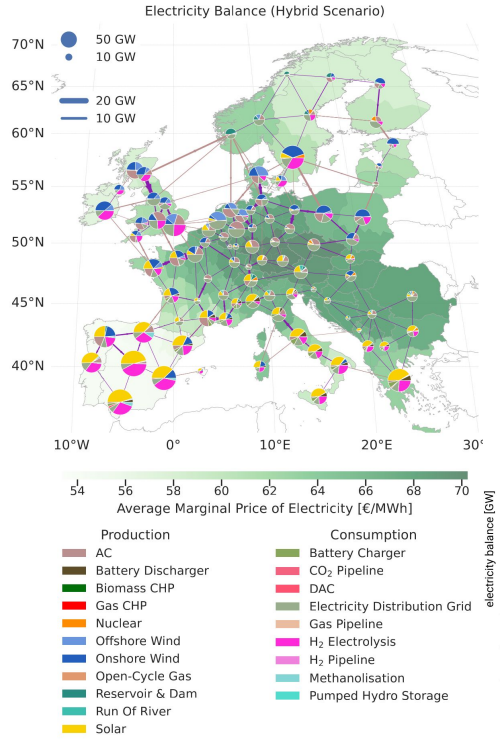
Backend

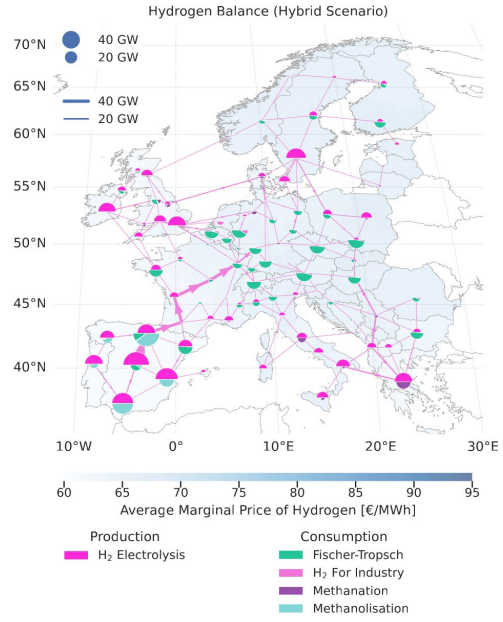
- all data stored in pandas
- framework built for performance with large networks and time series
- interfaces to major solvers (Gurobi, CPLEX, HiGHS, Xpress), with linopy (by PyPSA devs)
- highly customisable, but no GUI
- Suitable for greenfield, brownfield & pathway studies

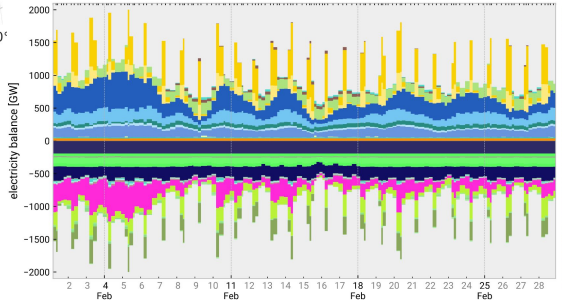
PyPSA-Eur: A sector-coupled open model of the European energy system

Automated workflow to build energy system model of Europe from raw open data with high spatial and temporal resolution:

1. OSM transmission lines (>220 kV) + TYNDP
2. a database of existing power plants,
3. time series for electricity demand,
4. time series for wind/solar availability, and
5. geographic wind/solar potentials
6. cost and efficiency assumptions
7. methods for model simplification
8. more for sector-coupled networks like pipelines, LNG terminals, electric vehicles, industry locations,







Energy infrastructure planning in PyPSA as an optimisation problem

Find the long-term cost-optimal energy system, including investments and short-term costs:

$$\textbf{Min}\left[\begin{array}{c}\textbf{Yearly}\\ \textbf{system costs}\end{array}\right] = \textbf{Min}\left[\sum_{n}\left(\begin{array}{c}\textbf{Annualised}\\ \textbf{capital costs}\end{array}\right) + \sum_{n,t}\left(\begin{array}{c}\textbf{Marginal}\\ \textbf{costs}\end{array}\right)\right]$$

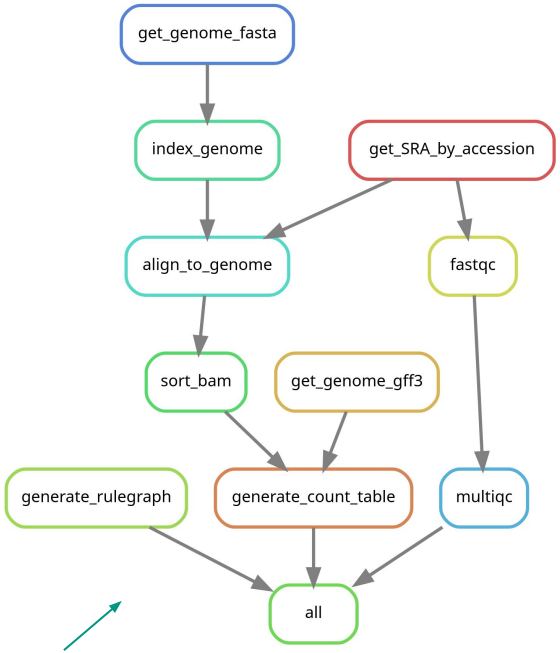
subject to

- meeting energy demand at each node n (e.g. region) and time t (e.g. hour of year) ← data
- transmission constraints between nodes and linearised power flow ← data
- wind, solar, hydro (variable renewables) availability time series $\forall\, n, t$ ← data
- installed capacity $\leq$ geographical potentials for renewables ← data
- fulfilling CO_2 emission reduction targets
- Flexibility from gas turbines, battery/hydrogen storage, HVDC links — More on that later!

Challenges with data-driven modelling

Create a full pipeline of data processing from raw data to results.

- Many different data sources
- Many data sources need cleaning and processing
- Many intermediate scripts and datasets
- Data and software dependencies need to be managed
- Data and code change over time
- Want to be able to reproduce results
- Want to run many different scenarios



Requires a scalable workflow management tool! → snakemake

Originally comes from bioinformatics field.

Miniature example of snakemake

Snakefile

```
rule mytask:
    input:
        "data/{sample}.txt"
    output:
        "result/{sample}.txt"
    script:
        "scripts/mytask.py"
```

```
rule myplot:
    input:
        "result/{sample}.txt"
    output:
        "figures/{sample}.pdf"
    script:
        "scripts/myplot.py"
```

command:

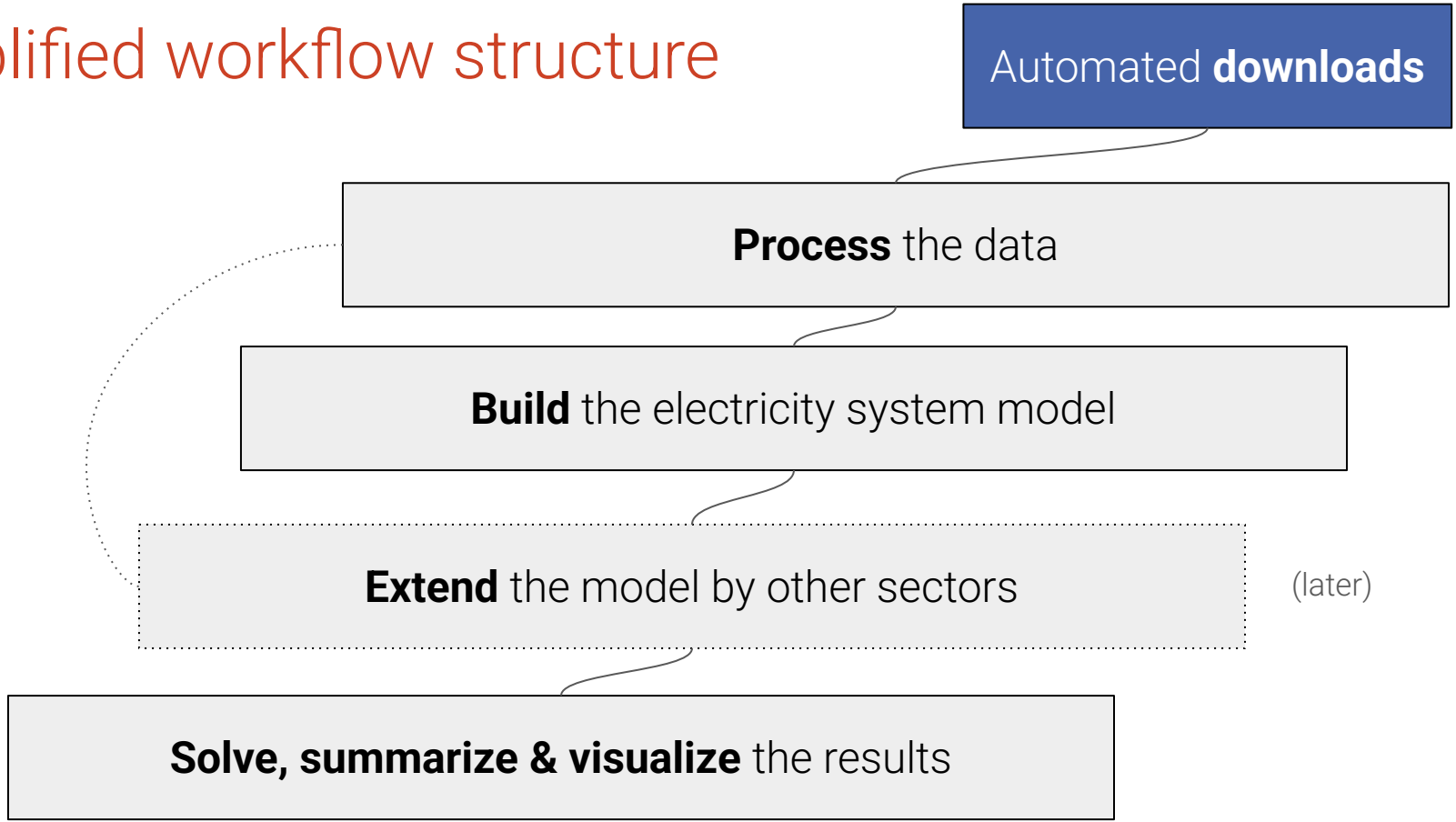
```
$ pixi run snakemake figures/myfigure.pdf
```

Example adapted from Johannes Köster, https://slides.com/johanneskoester/snakemake-short#/



snakemake workflow for the electricity sector

Simplified workflow structure

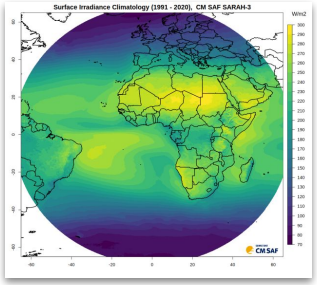


First, raw data is automatically downloaded.

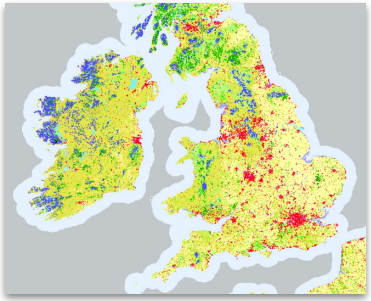
WDPA



SARAH-3



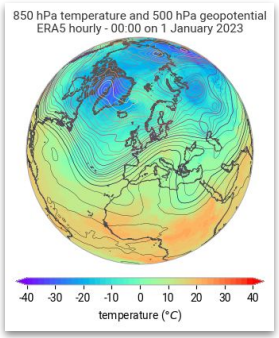
CORINE



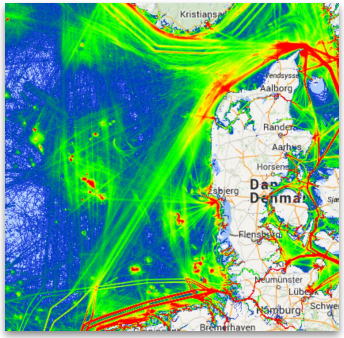
GEBCO

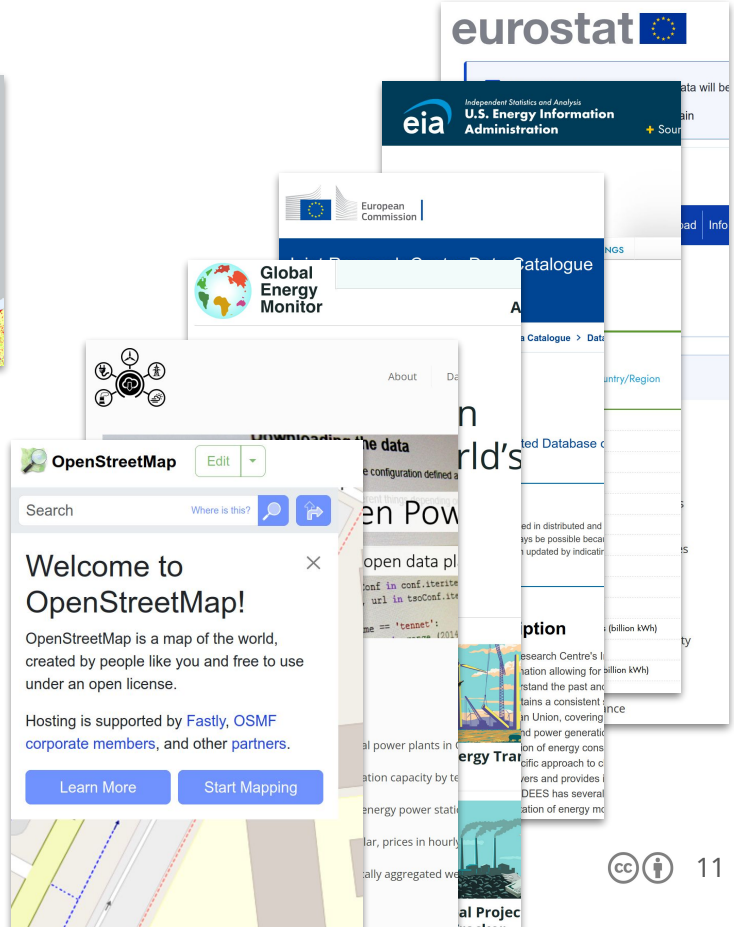


ERA5



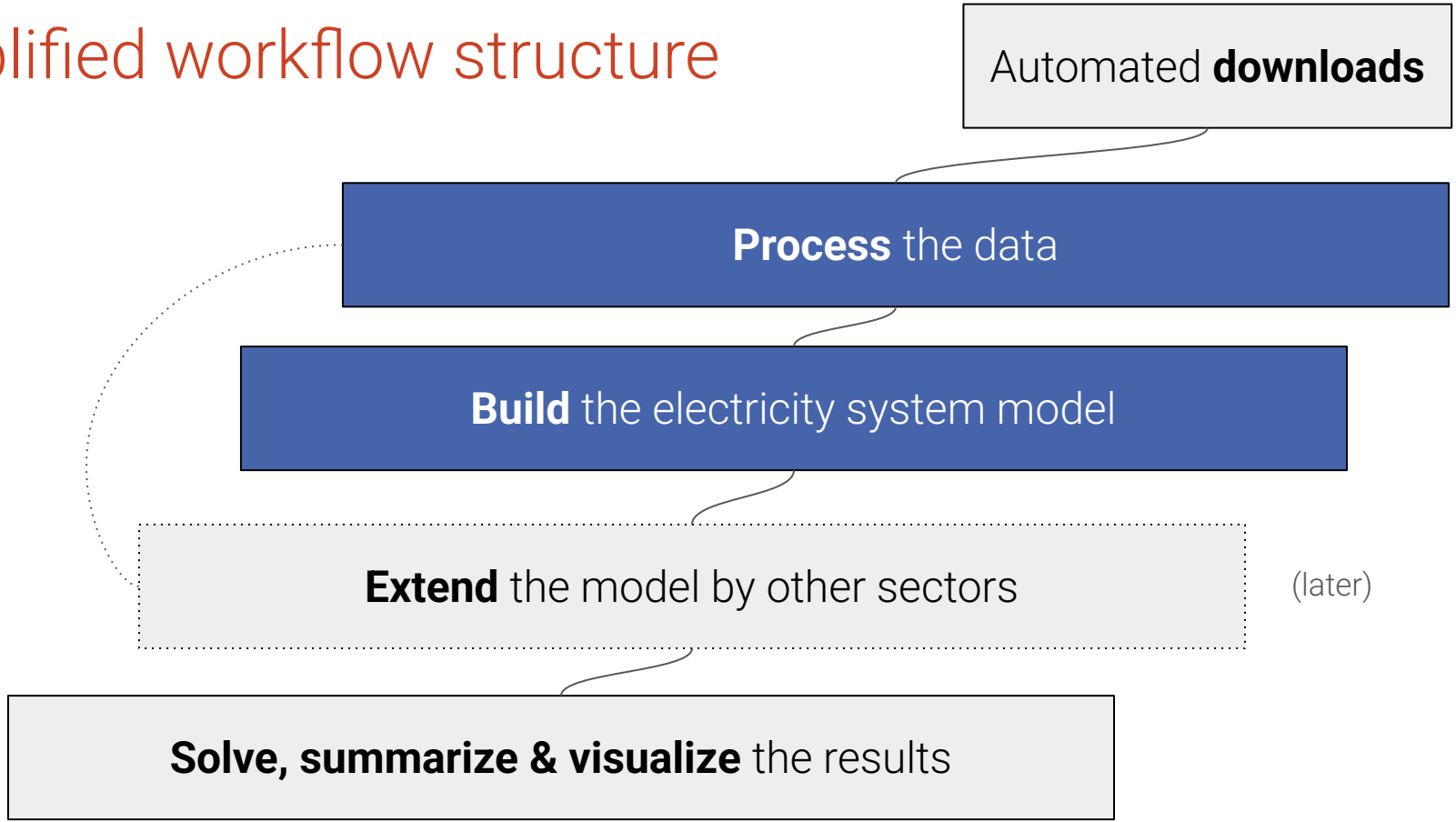
World Bank





https://pypsa-eur.readthedocs.io/en/latest/data_sources.html

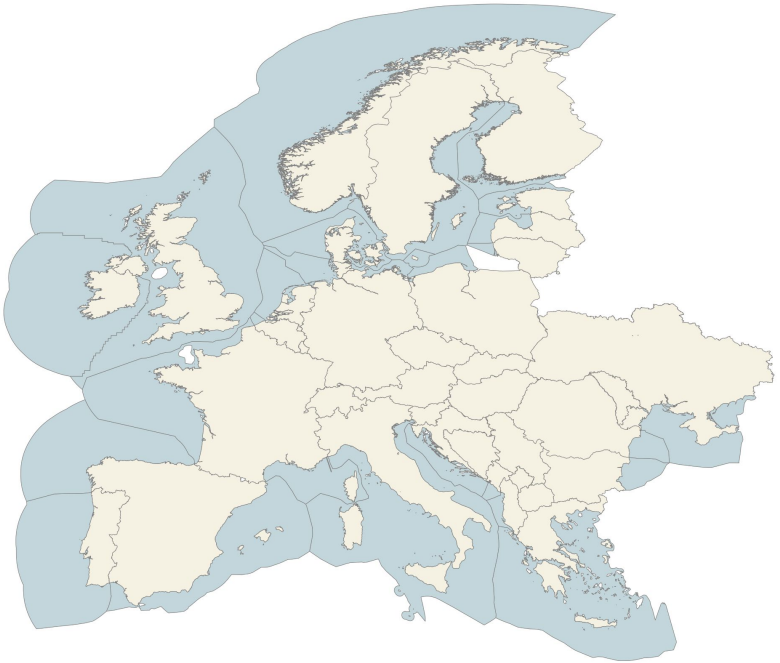
Simplified workflow structure



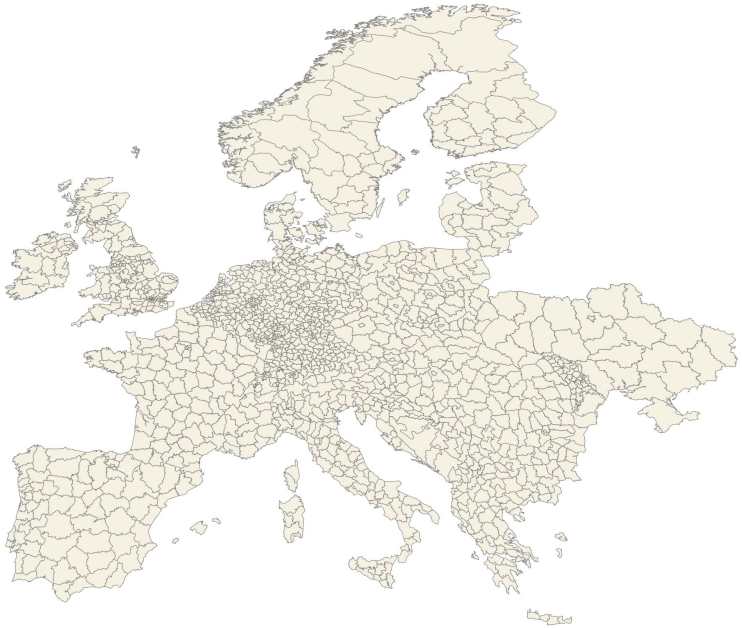
Steps to building PyPSA-Eur electricity system

Retrieve onshore & offshore polygons for each country	`build_shapes`

Country shapes & exclusive economic zones (EEZ)



NUTS administrative regions (NUTS3)

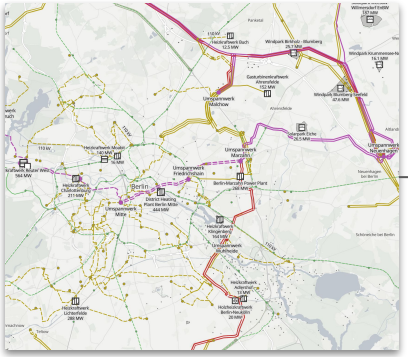


Steps to building PyPSA-Eur electricity system

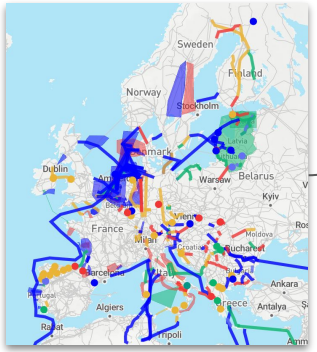
Retrieve onshore & offshore polygons for each country	`build_shapes`
Construct a base high-voltage network with buses, transformers, AC & DC lines with DLR & TYNDP	`base_network`, `build_transmission_projects`

Power grid topology

OpenStreetMap data



TYNDP projects



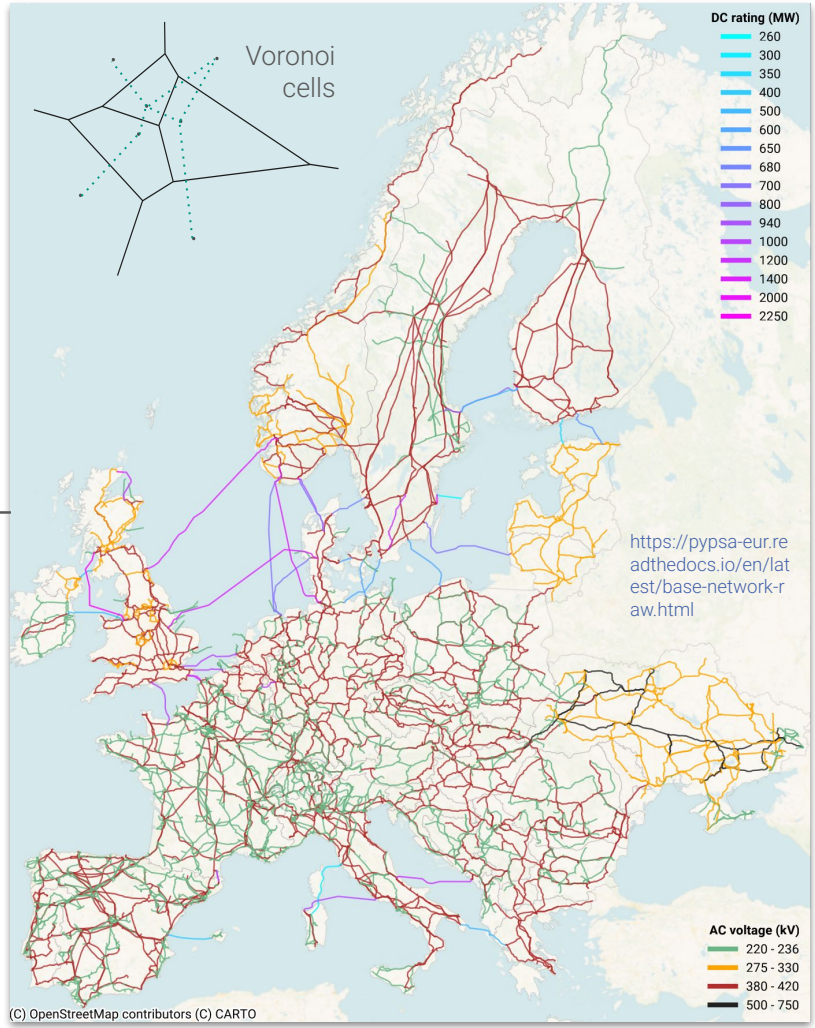
Apply **standard line types** for capacity and parameters.

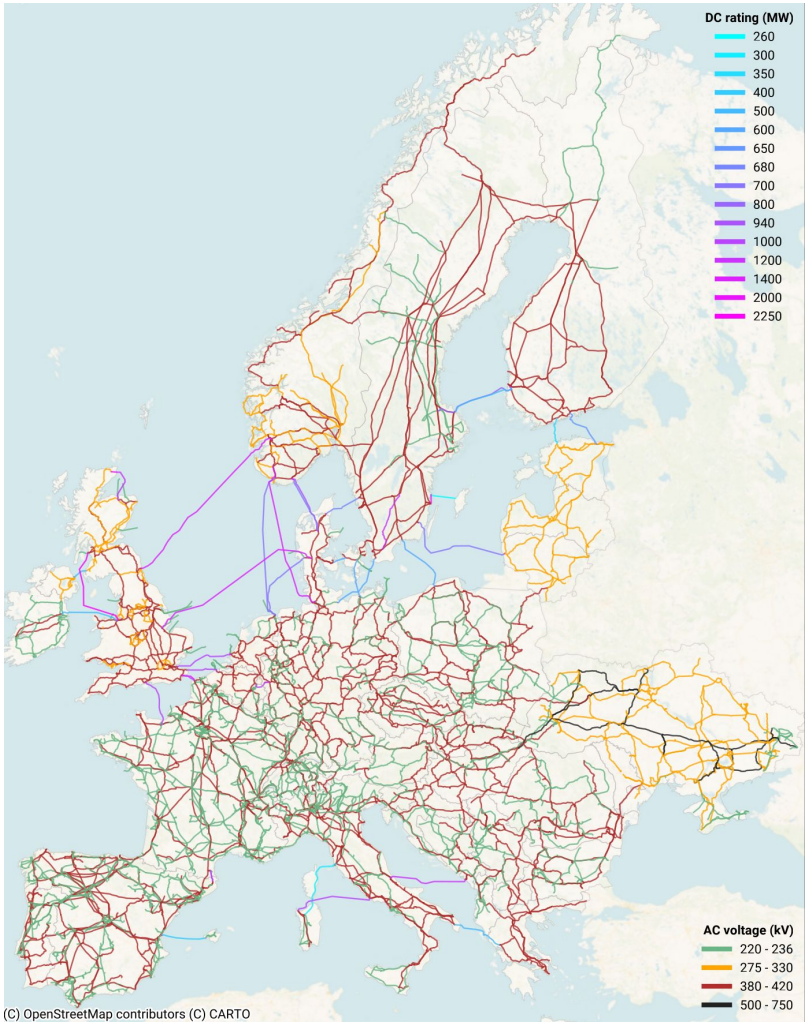
Calculate **dynamic line rating** potential from weather data.

European network with

- ~5,800 buses
- ~7,300 AC lines (>220 kV)
- 36 HVDC links (+TYNDP)

https://www.nature.com/articles/s41597-025-04550-7




DC rating (MW)
260
300
350
400
500
600
650
680
700
800
940
1000
1200
1400
2000
2250
AC voltage (kV)
220 - 236
275 - 330
380 - 420
500 - 750
(C) OpenStreetMap contributors (C) CARTO

Steps to building PyPSA-Eur electricity system

Retrieve onshore & offshore polygons for each country	`build_shapes`
Construct a base high-voltage network with buses, transformers, AC & DC lines with DLR & TYNDP	`base_network`, `build_transmission_projects`
Transform all transmission lines to 380kV, remove dead ends & cluster with k-means or hierarchical clustering	`simplify_network`, `cluster_network`

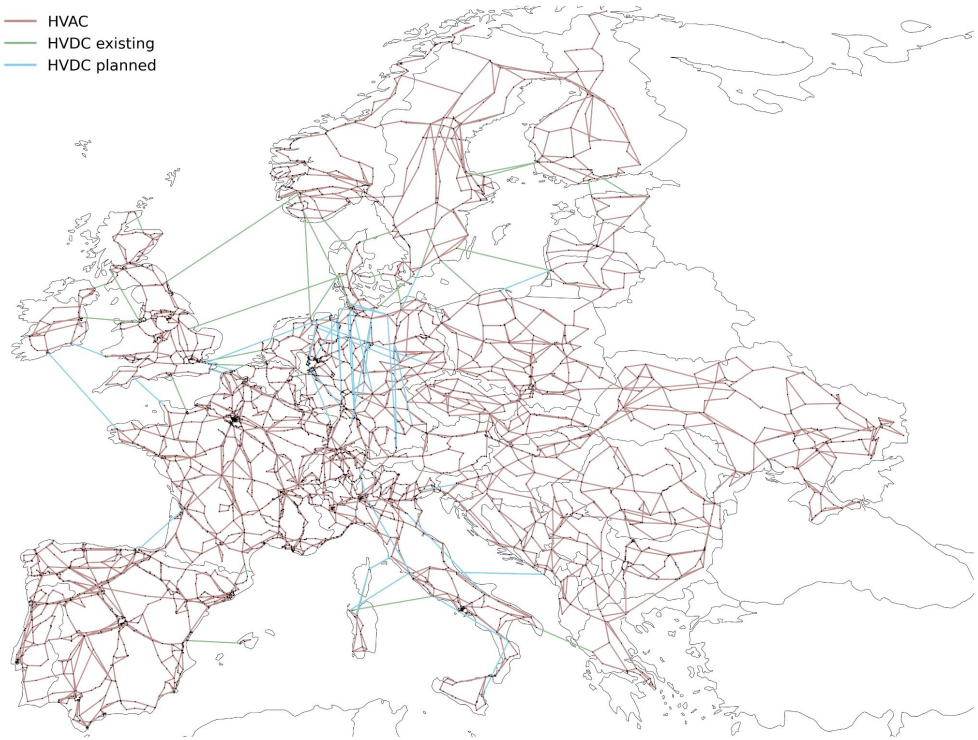
Clustering the electricity network: `simplify_network`

Need to make the optimization problem less **computationally challenging**…

…if we want to **co-optimize** generation, storage, PtX conversion and transmission infrastructure:

1. Lift all lines to common voltage level of 380 kV.

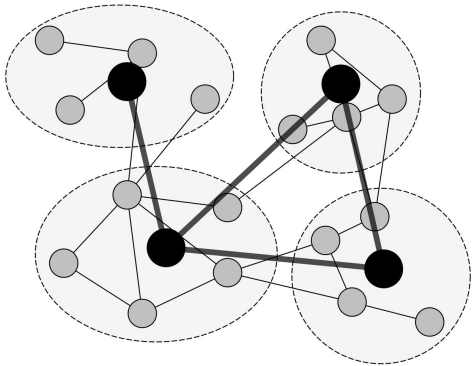
2. Remove dead ends.

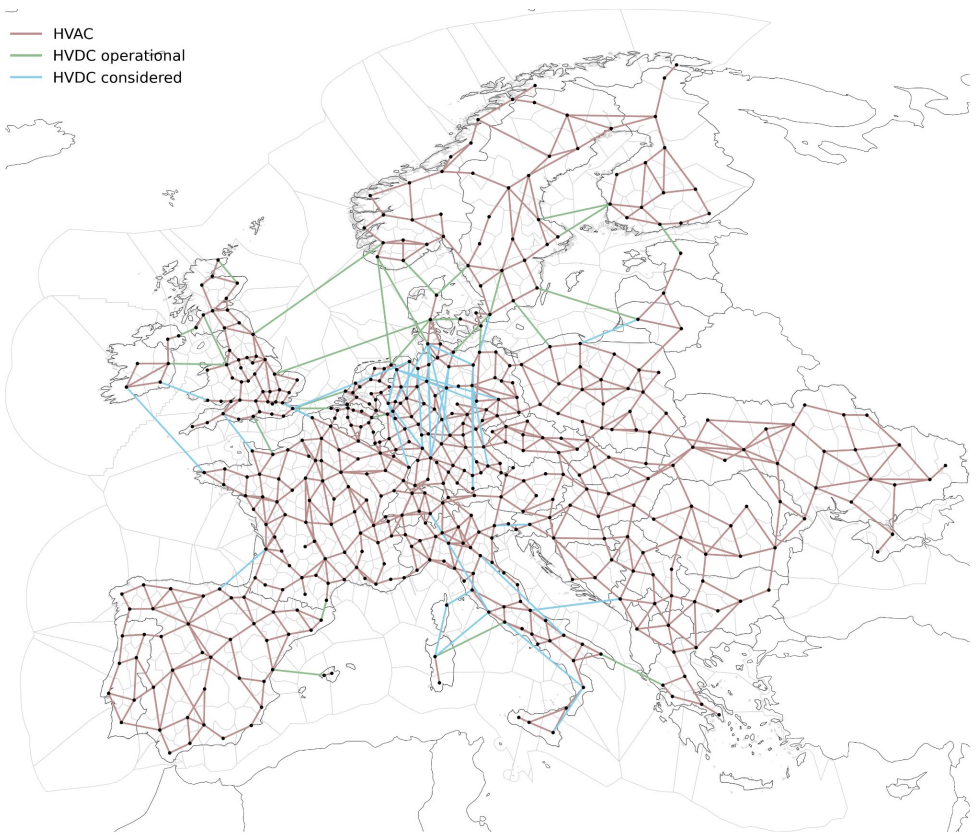


Clustering the electricity network: `cluster_network`

Transformed **to 380 kV**

Clustered to **512 regions**

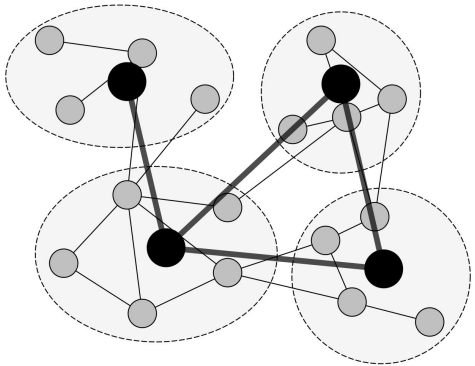


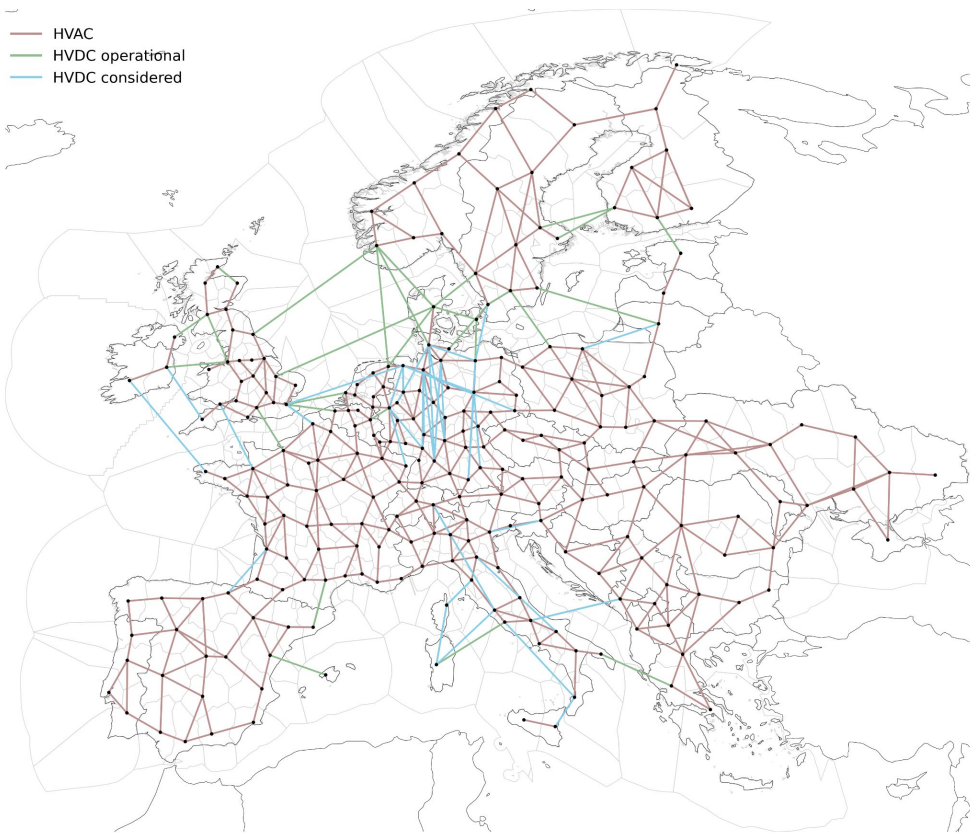


Clustering the electricity network: `cluster_network`

Transformed **to 380 kV**

Clustered to **256 regions**

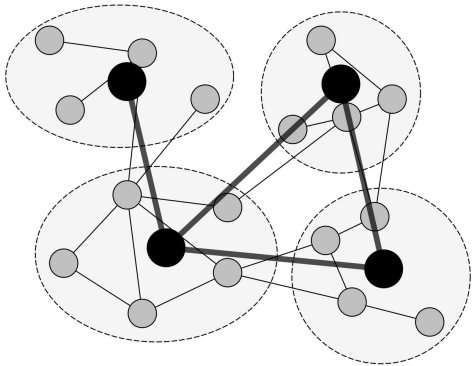




Clustering the electricity network: `cluster_network`

Transformed **to 380 kV**

Clustered to **128 regions**

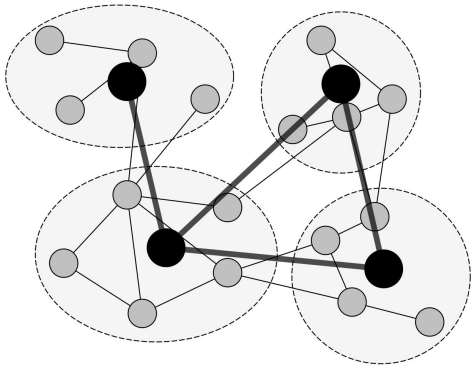




Clustering the electricity network: `cluster_network`

Transformed **to 380 kV**

Clustered to **64 regions**

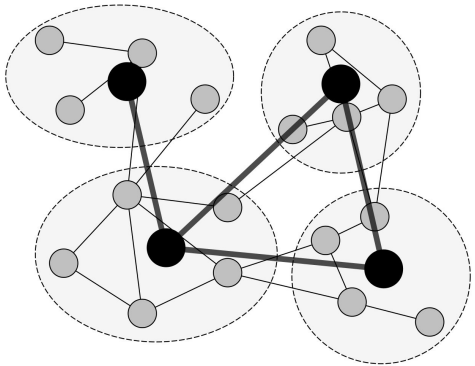




Clustering the electricity network: `cluster_network`

Transformed **to 380 kV**

Clustered to **41 regions**





Steps to building PyPSA-Eur electricity system

Retrieve onshore & offshore polygons for each country	`build_shapes`
Construct a base high-voltage network with buses, transformers, AC & DC lines with DLR & TYNDP	`base_network`, `build_transmission_projects`
Transform all transmission lines to 380kV, remove dead ends & cluster with k-means or hierarchical clustering	`simplify_network`, `cluster_network`
Determine eligible areas for utility-scale PV & onshore/offshore wind park development	`determine_availability_matrix`
Build renewable capacity factor profiles for each clustered region based on land availability	`build_renewable profiles`, `build_hydro_profile`

atlite: Convert weather data to energy systems data

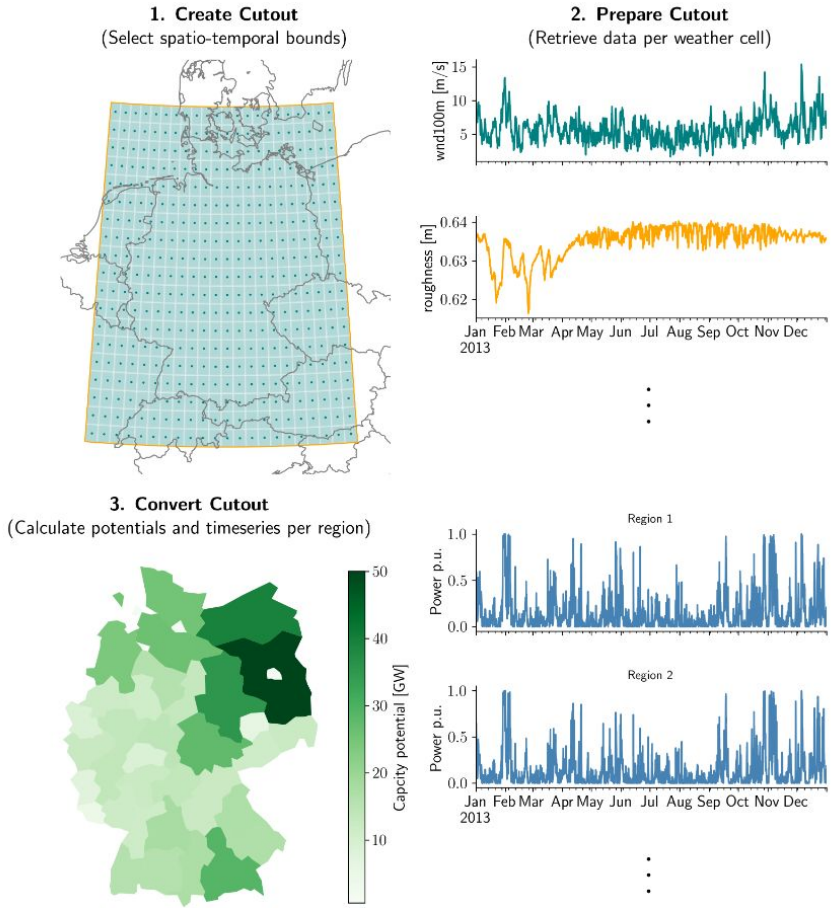
Rule: `build_renewable profiles`

pypi v0.3.0 | conda-forge v0.3.0 | Tests passing | codecov 72% | docs passing | license MIT
REUSE compliant | JOSS 10.21105/joss.03294 | chat 52 online | stackoverflow pypsa questions 44

Python library for converting weather data (e.g. wind, solar radiation, temperature, precipitation) into energy systems data:

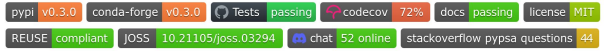
- solar photovoltaics
- solar thermal collectors
- wind turbines
- hydro run-off, reservoir, dams
- heat pump COPs
- dynamic line rating (DLR)
- heating and cooling demand (HDD/CDD)

It can also perform land eligibility analyses.

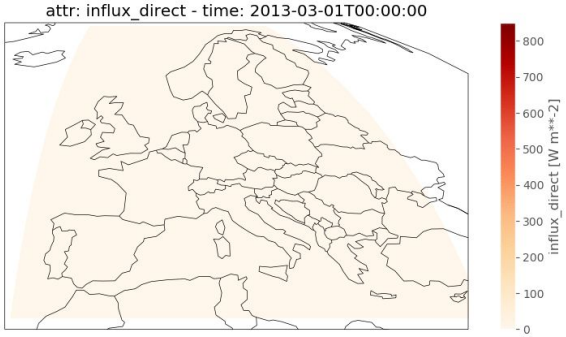


Time series for renewables

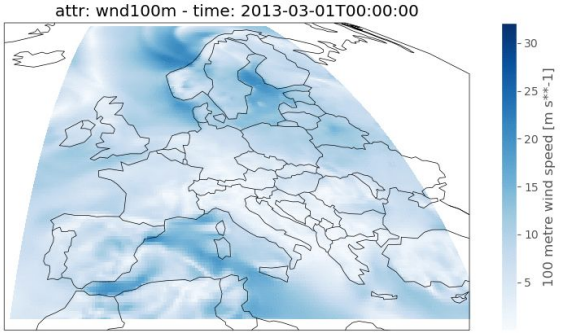
atlite: Convert weather data to energy systems data



Historical meteorological weather data from ERA5 and SARAH-3
(up to 84 years, 30x30 km)

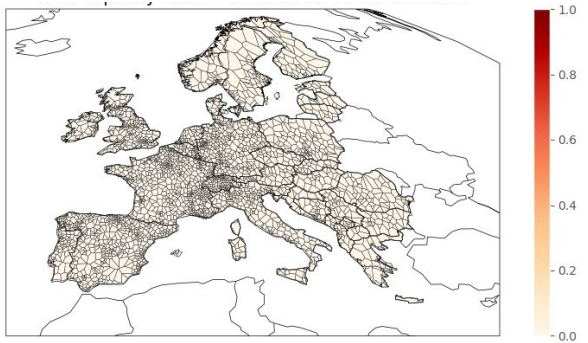


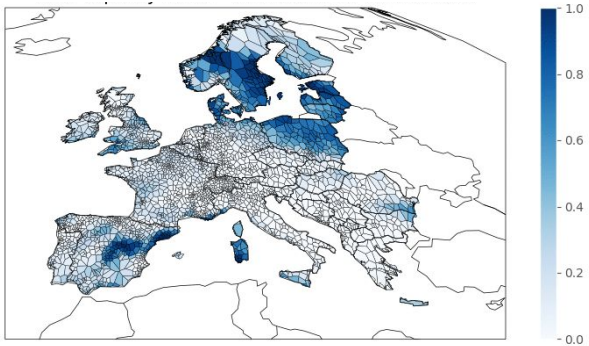
Solar panel models
- orientation
- material



Wind turbine models
- power curve
- surface roughness

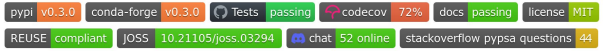
Wind and solar capacity factors





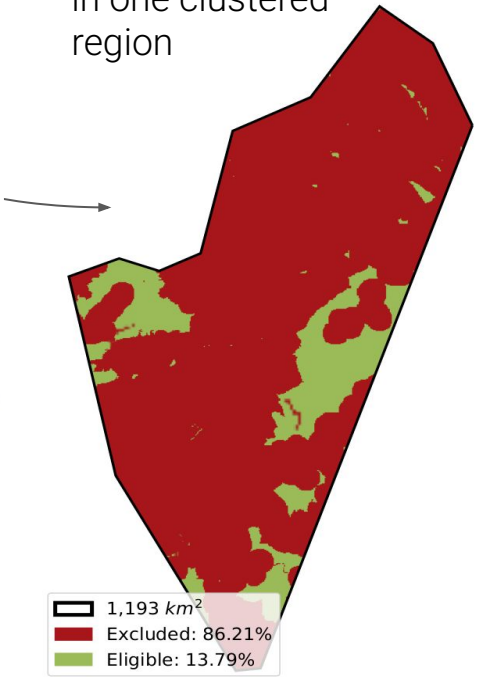
Land availability for renewables

atlite: Convert weather data to energy systems data





Example:
Onshore wind
in one clustered
region



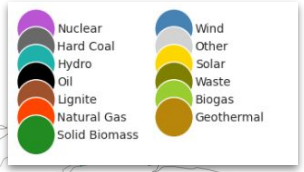
- CORINE / LUISA land cover
  - eligible land types
  - distance requirements
- NATURA / WDPA natural protection areas
- GEBCO bathymetry data
- Shipping lanes
- Distance to shore

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Transform all transmission lines to 380kV, remove dead ends & cluster with k-means or hierarchical clustering	`simplify_network`, `cluster_network`
Determine eligible areas for utility-scale PV & onshore/offshore wind park development	`determine_availability_matrix`
Build renewable capacity factor profiles for each clustered region based on land availability	`build_renewable profiles`, `build_hydro_profile`
Prepare existing renewables and fossil power plants	`build_powerplants`

Welcome to powerplantmatching's documentation!

https://globalenergymonitor.org/projects/global-integrated-power-tracker/tracker-map/



pypi v0.7.0 | conda-forge v0.7.0 | python >=3.9 | Tests failing | docs passing | pre-commit.ci passed | Ruff

license GPLV3+ | DOI 10.5281/zenodo.3358985 | stackoverflow pypsa questions 44

A toolset for cleaning, standardizing and combining multiple power plant databases.

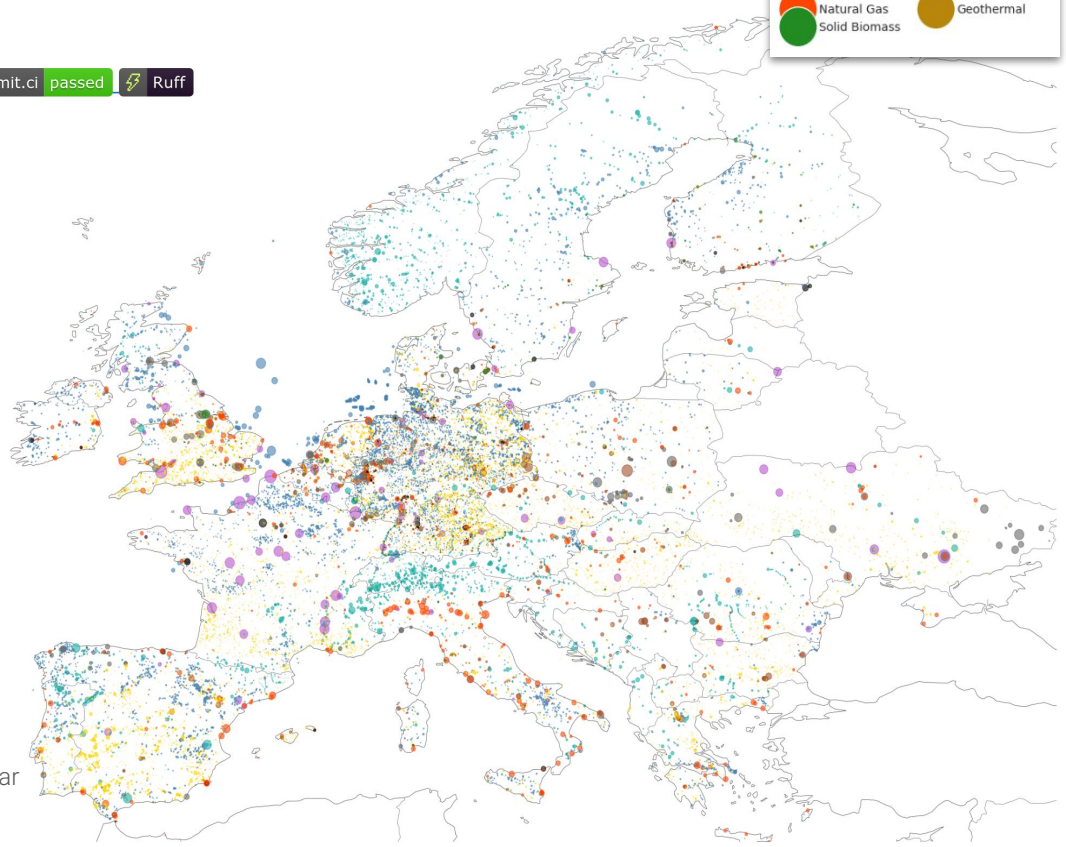
```
import powerplantmatching as pm
df = pm.powerplants(from_url=True)
df.query("DateIn > 2000")
```

Sources

- Global Energy Monitor (GEM)
- Open Power System Data (OPSD)
- Global Energy Observatory
- World Resources Institute
- Marktstammdatenregister (MaStR)
- CARMA
- ENTSO-E, BNetzA, UBA, IRENA
- JRC for hydro power plants

Attributes

- name
- fuel type
- technology
- country
- capacity
- commissioning year
- retirement year
- coordinates



github.com/pypsa/powerplantmatching

Steps to building PyPSA-Eur electricity system

Retrieve onshore & offshore polygons for each country	`build_shapes`
Construct a base high-voltage network with buses, transformers, AC & DC lines with DLR & TYNDP	`base_network`, `build_transmission_projects`
Transform all transmission lines to 380kV, remove dead ends & cluster with k-means or hierarchical clustering	`simplify_network`, `cluster_network`
Determine eligible areas for utility-scale PV & onshore/offshore wind park development	`determine_availability_matrix`
Build renewable capacity factor profiles for each clustered region based on land availability	`build_renewable profiles`, `build_hydro_profile`
Prepare existing renewables and fossil power plants	`build_powerplants`
Add generation, storage and demand to the network with techno-economic assumptions on costs and efficiencies, ...	`compose_network`

Open database of techno-economic assumptions

- compiles **techno-economic assumptions** on energy system components
  - investment costs, FOM/VOM costs, efficiencies, lifetimes
  - for given years, e.g. 2020, 2030, 2040, 2050
  - from mixed sources, but prioritising **Danish Energy Agency** where available (and sensible)

Preview | Code | Blame | 1097 lines (1097 loc) · 213 KB | Raw

fischer-tropsch

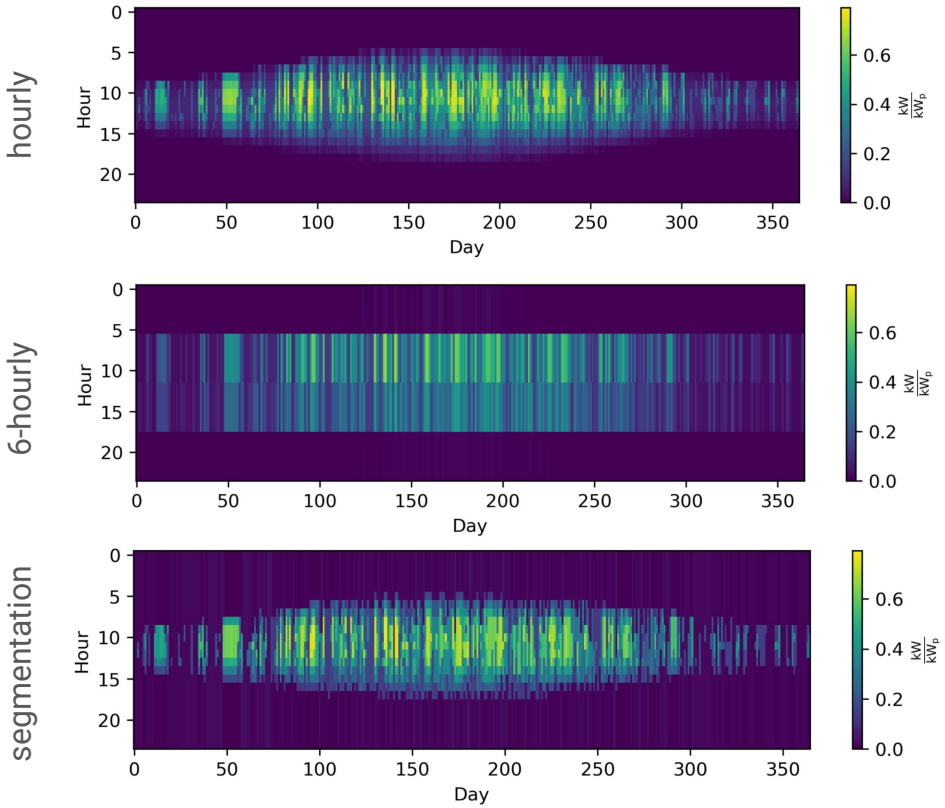
1	technology	parameter	value	unit	source	fu
217	Fischer-Tropsch	FOM	3.0	%/year	Agora Energiewende (2018): The Future Cost of Electricity-Based Synthetic Fuels (https://www.agora-energiewende.de/en/publications/the-future-cost-of-electricity-based-synthetic-fuels-1/), section 6.3.2.1.	
218	Fischer-Tropsch	VOM	4.4663	EUR/MWh_FT	Danish Energy Agency, data_sheets_for_renewable_fuels.xlsx	1(
219	Fischer-Tropsch	capture rate	0.9	per unit	Assumption based on doi:10.1016/j.biombioe.2015.01.006	
220	Fischer-Tropsch	carbondioxide-input	0.326	t_CO2/MWh_FT	DEA (2022): Technology Data for Renewable Fuels (https://ens.dk/en/our-services/projections-and-models/technology-data/technology-data-renewable-fuels), Hydrogen to Jet Fuel, Table 10 / pg. 267.	Ir
221	Fischer-Tropsch	efficiency	0.799	per unit	Agora Energiewende (2018): The Future Cost of Electricity-Based Synthetic Fuels (https://www.agora-energiewende.de/en/publications/the-future-cost-of-electricity-based-synthetic-fuels-1/), section 6.3.2.2.	
222	Fischer-Tropsch	electricity-input	0.007	MWh_el/MWh_FT	DEA (2022): Technology Data for Renewable Fuels (https://ens.dk/en/our-services/projections-and-models/technology-data/technology-data-renewable-fuels), Hydrogen to Jet Fuel, Table 10 / pg. 267.	0.
223	Fischer-Tropsch	hydrogen-input	1.421	MWh_H2/MWh_FT	DEA (2022): Technology Data for Renewable Fuels (https://ens.dk/en/our-services/projections-and-models/technology-data/technology-data-renewable-fuels), Hydrogen to Jet Fuel, Table 10 / pg. 267.	0.
224	Fischer-Tropsch	investment	703726.4462	EUR/MW_FT	Agora Energiewende (2018): The Future Cost of Electricity-Based Synthetic Fuels (https://www.agora-energiewende.de/en/publications/the-future-cost-of-electricity-based-synthetic-fuels-1/), table 8: "Reference scenario".	W
225	Fischer-Tropsch	lifetime	20.0	years	Danish Energy Agency, Technology Data for Renewable Fuels (04/2022), Data sheet "Methanol to Power".	
956	methanation	lifetime	20.0	years	Guesstimate.	B

`https://github.com/PyPSA/technology-data/blob/master/outputs/costs_2030.csv`

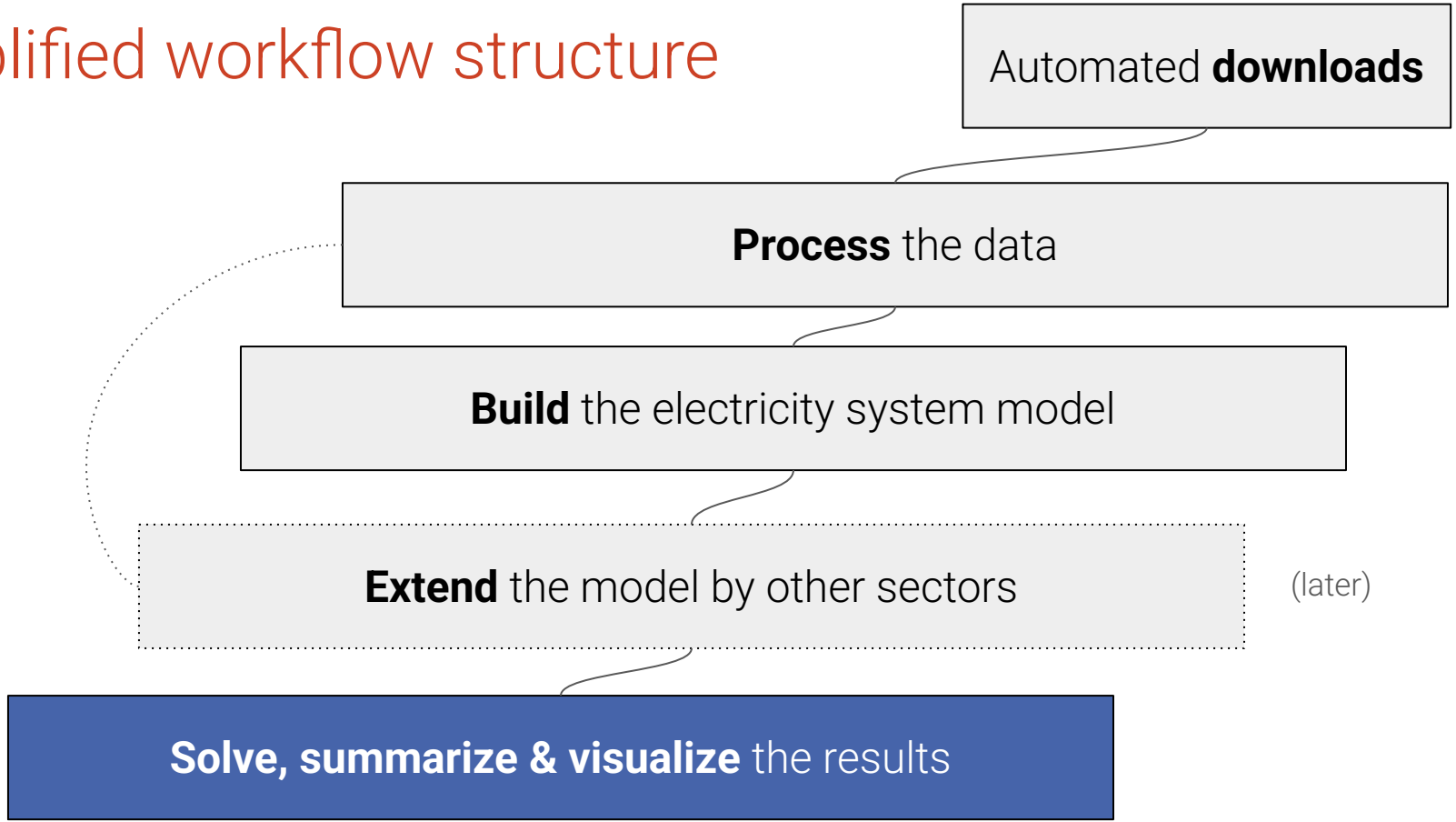
Temporal aggregation

Multiple options:

1. **averaging** of every Nth hour
2. **sampling** every Nth hour (e.g. 3-hourly)
3. Non-equidistant **segmentation** with pre-defined number of segments using the `tsam` Python library from **FZ Jülich**



Simplified workflow structure



linopy: Linear optimization with N-D labeled variables



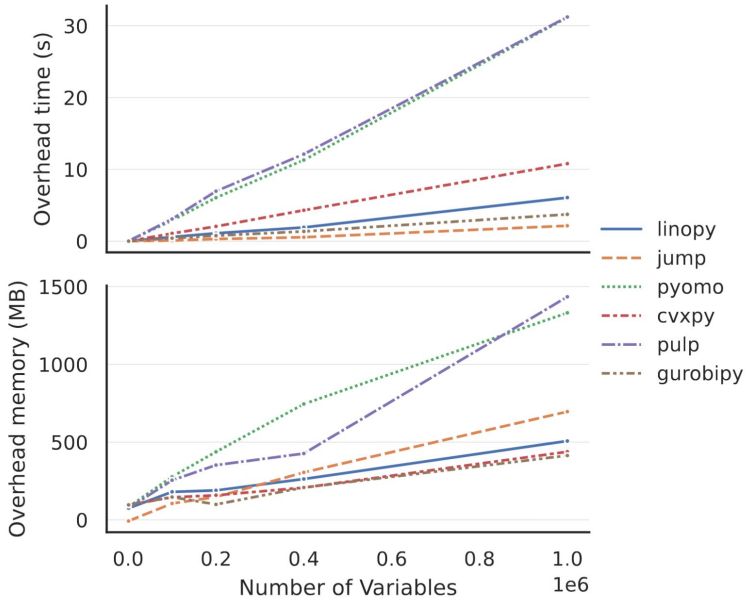


Python library that facilitates optimization with real-world, large-scale data.

It supports:

- Linear (LP),
- Mixed-Integer (MILP),
- Quadratic programming (QP).

It has been developed to make linear programming in Python easy, highly-flexible and – most importantly – highly performant.



https://joss.theoj.org/papers/10.21105/joss.04823

Solving and summarising networks

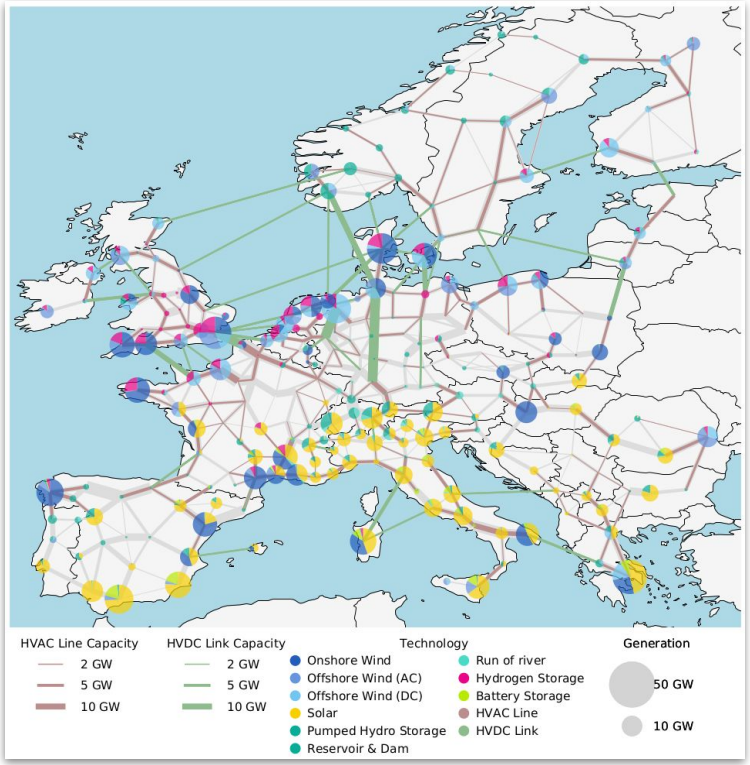
Hardware requirements:

- Building the model **can run locally** on most modern laptops. Very simple models can run with HiGHS solver.

- But access to a **commercial solver** and a larger **cluster/workstation** is required for solving problems (~250 GB RAM per scenario if resolution is very high)!

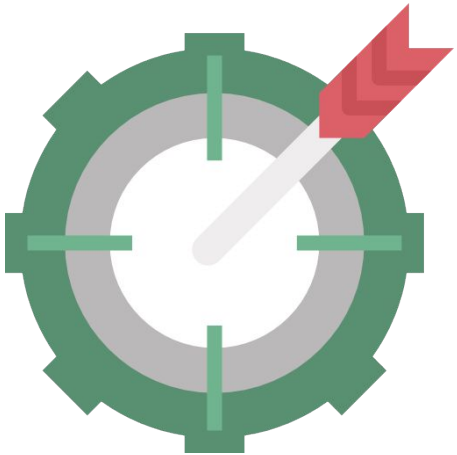
There is a **statistics module in PyPSA** designed to help with analysing solved networks and several **figures/maps are created automatically**.

Example result without sector-coupling



What is configurable?

electricity-only examples

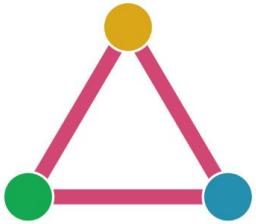


`config.yaml`

- Select subset of countries and focus countries (e.g. only DE)
- Select weather year (1940 - 2024 for ERA5)
- Specify CO_2 constraint and gas usage limit
- Tweak spatial resolution (between 41 and >1000 nodes)
- Tweak temporal resolution (from hourly to N-hourly)
- Customize cost assumptions (e.g. 2020, 2030, 2050)
- Parametrize technologies (e.g. wind turbine type, panel orientation)
- Define land use eligibility criteria (e.g. distance requirements)
- Pick a solver (HiGHS, Gurobi, CPLEX, Xpress…)
- Choose between greenfield or brownfield expansion

Let's look at this in more detail!

Icon made by surang from www.flaticon.com



Search Ctrl + K

Getting Started

- Introduction
- Installation
- Tutorial: Electricity-Only
- Tutorial: Sector-Coupled

Configuration

- Wildcards
- **Configuration**
- Foresight Options
- Techno-Economic Assumptions

Rules Overview

- Retrieving Data
- Building Electricity Networks
- Building Sector-Coupled Networks
- Solving Networks
- Plotting and Summaries

Configuration

PyPSA-Eur has several configuration options which are documented in this section and are collected in a `config/config.yaml` file. This file defines deviations from the default configuration (`config/config.default.yaml`); confer installation instructions at Handling Configuration Files.

Top-level configuration

"Private" refers to local, machine-specific settings or data meant for personal use, not to be shared. "Remote" indicates the address of a server used for data exchange, often for clusters and data pushing/pulling.

```yaml
version: v2025.01.0
tutorial: false

logging:
  level: INFO
  format: '%(levelname)s:%(name)s:%(message)s'

private:
  keys:
    entsoe_api:

remote:
  ssh: ""
  path: ""
```

	Unit	Values	Description
version	–	0.x.x	Version of PyPSA-Eur. Descriptive only.
tutorial	bool	{true, false}	Switch to retrieve the tutorial data set instead of the full data set.
logging			

Contents

Top-level configuration

- `run`
- `foresight`
- `scenario`
- `countries`
- `snapshots`
- `enable`
- `co2 budget`
- `electricity`
- `atlite`
- `renewable`
- `conventional`
- `lines`
- `links`
- `transmission projects`
- `transformers`
- `load`
- `energy`
- `biomass`
- `solar_thermal`
- `existing_capacities`
- `sector`
- `industry`
- `costs`
- `clustering`

https://pypsa-eur.readthedocs.io/en/latest/configuration/

Live Demo – Belgium / electricity-only / few days

Start with a dry-run:

```
$ pixi run snakemake -c1 --configfile config/test/config.electricity.yaml -n
```

Then execute the same command “for real” by dropping “-n” flag:

The “-c1” flag tells snakemake to run one job at a time; “-call” to use all cores.

```
$ pixi run snakemake -c1 --configfile config/test/config.electricity.yaml
```

To explore results, start a Jupyter notebook:

```
$ pixi run jupyter notebook
```

Practical Phase

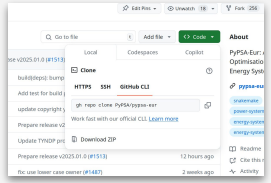
(electricity-only)

1) Download the repository

Open a terminal and type:

```
$ cd ~/path/to/my/directory
$ git clone https://github.com/PyPSA/pypsa-eur.git
$ cd pypsa-eur
$ git checkout v2026.09.0
```

You can also download the repository as a ZIP by hand.



2) Install pixi following instructions

https://pixi.prefix.dev/latest/installation/

3) Run PyPSA-Eur tutorial with `snakemake`

Guide:

https://pypsa-eur.readthedocs.io/en/latest/tutorial/

```
$ pixi run snakemake -call --configfile
  config/test/config.electricity.yaml
```

Users of Windows, add two lines to YAML:

```
run:
  use_shadow_directory: false
```

4) Explore PyPSA network in a Jupyter notebook

```
import pypsa
fn = "results/test-elec/networks/solved_2050.nc"
n = pypsa.Network(fn)
n.statistics()
n.explore()
```

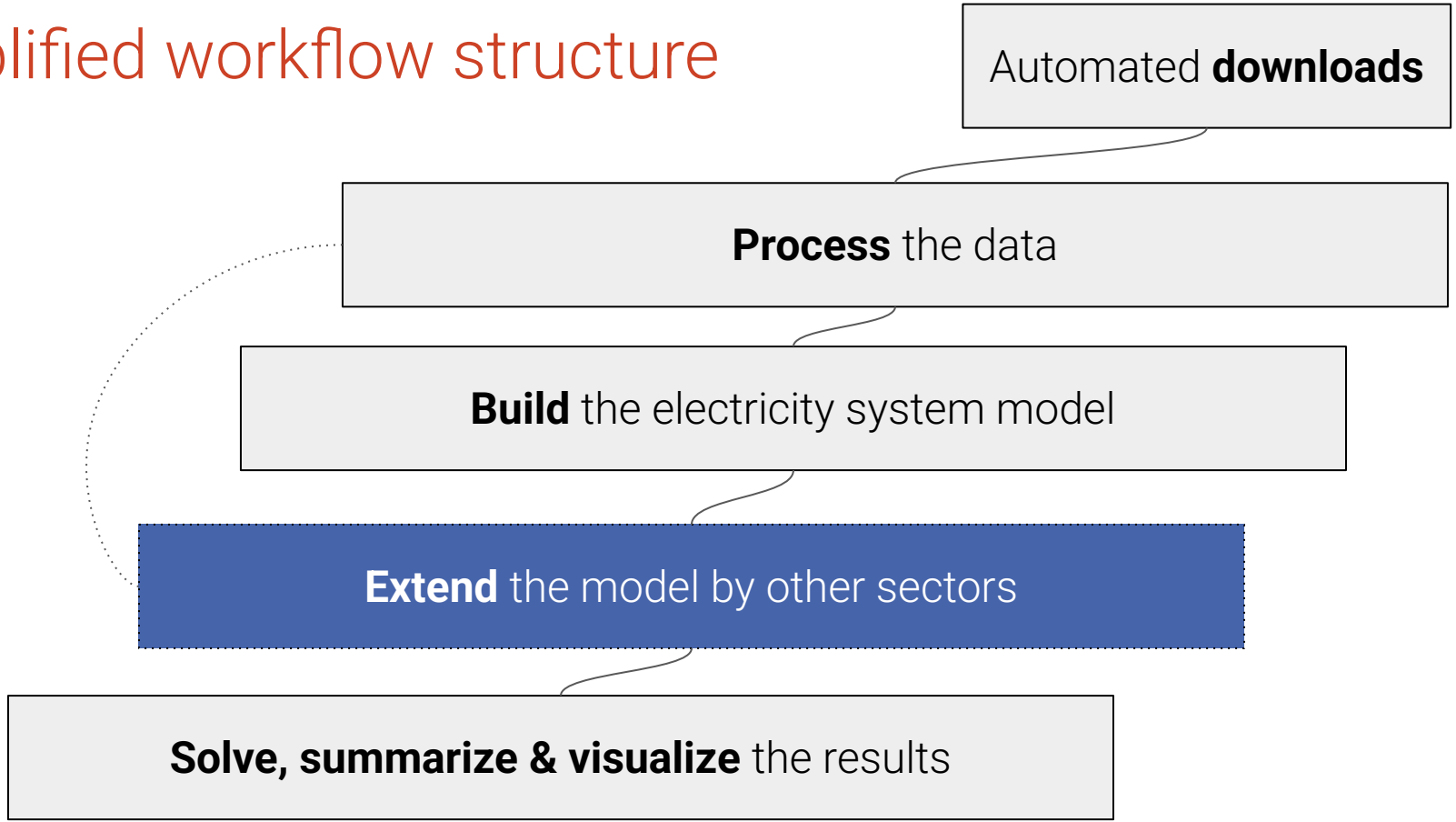
Small exploratory configuration tasks

(electricity-only)

Go to https://pypsa-eur.readthedocs.io/en/latest/configuration and https://github.com/PyPSA/pypsa-eur/blob/master/config/config.default.yaml and try to find out how to configure some of the settings for **electricity-only models** listed below:

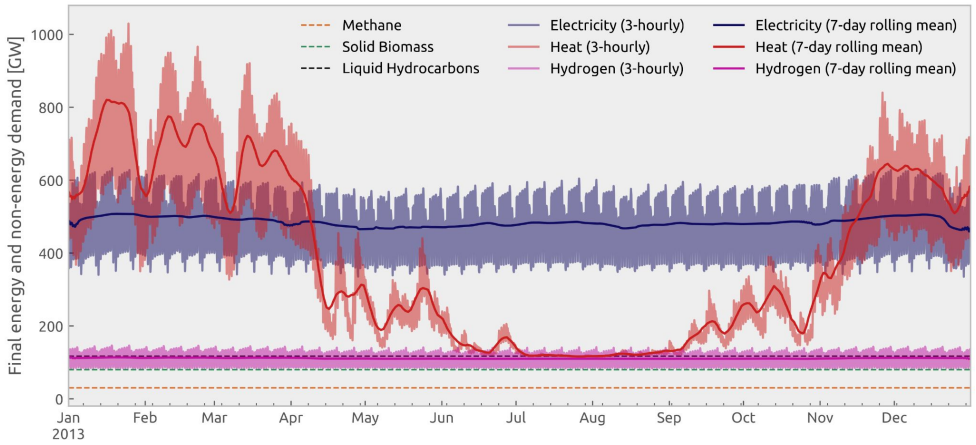
1. Increase the maximum line loading from 70% to 100%.
2. Disable power transmission grid reinforcements.
3. Activate dynamic line rating with default settings.
4. Activate linearised transmission loss approximation.
5. Deactivate the estimation of existing renewable capacities.
6. Change the techno-economic assumptions to the year 2020.
7. Remove the option to build hydrogen or battery storage.

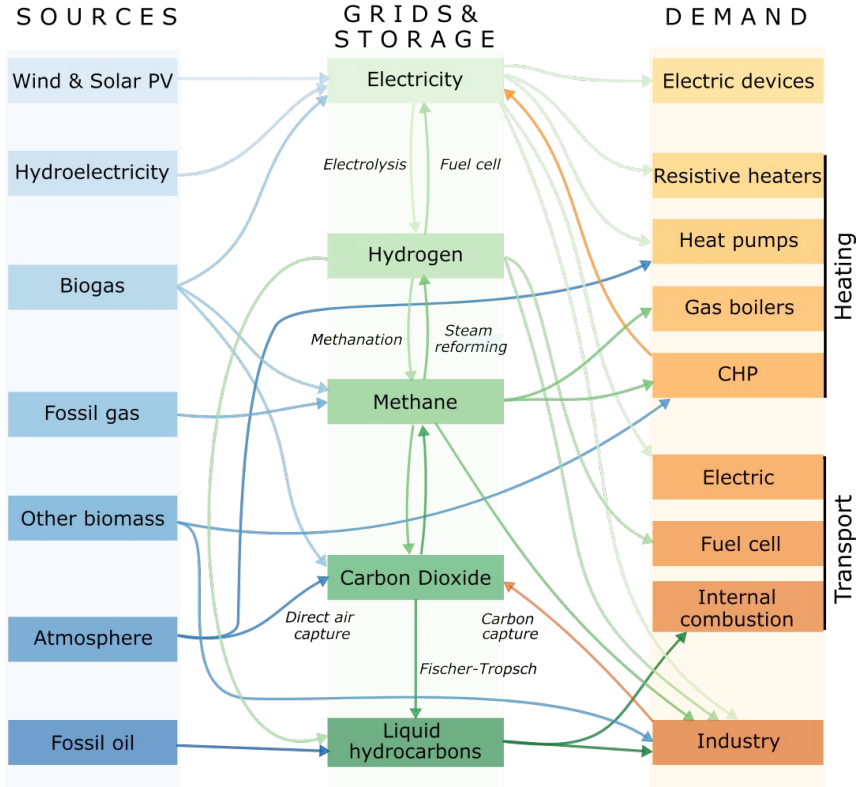
Simplified workflow structure



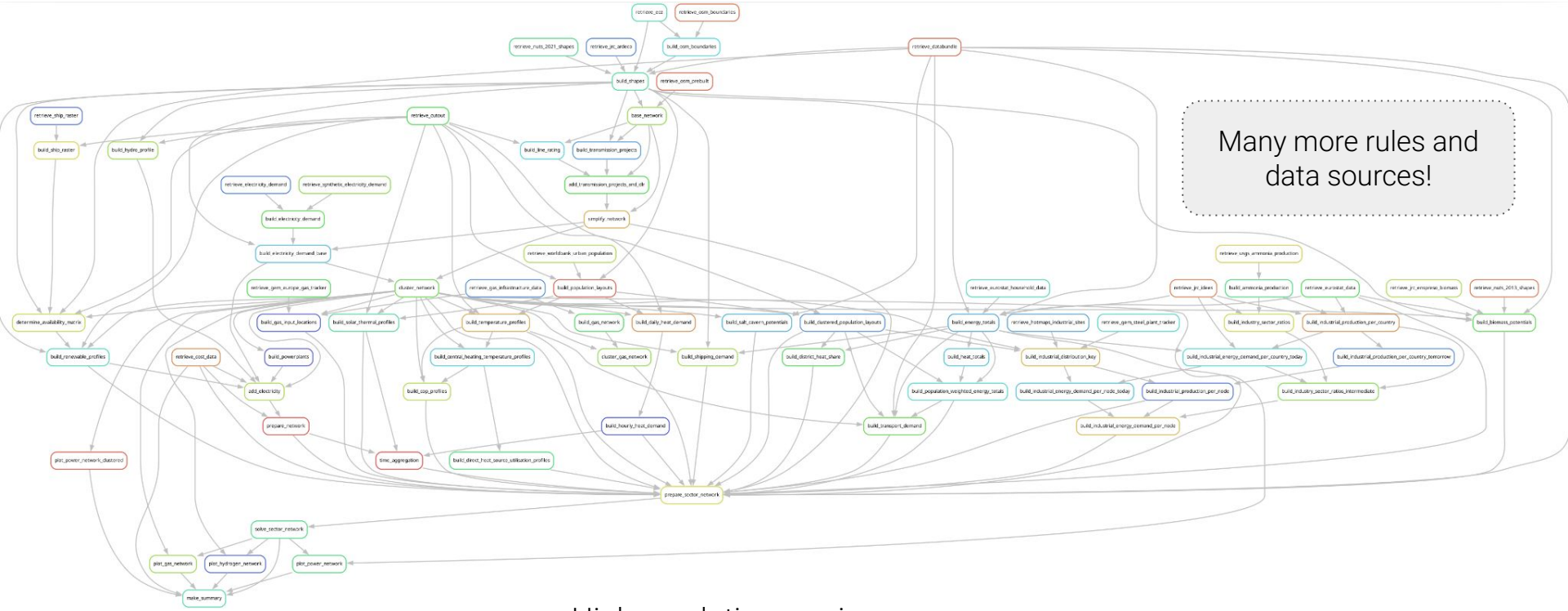
Coupling with other sectors

- transport sector (EVs, shipping, aviation)
- heating sector (district heating, individual)
- industry sector (steel, chemicals, ammonia, …)
- carbon management (CCUTS) + biomass
- hydrogen, CO_2 and gas networks





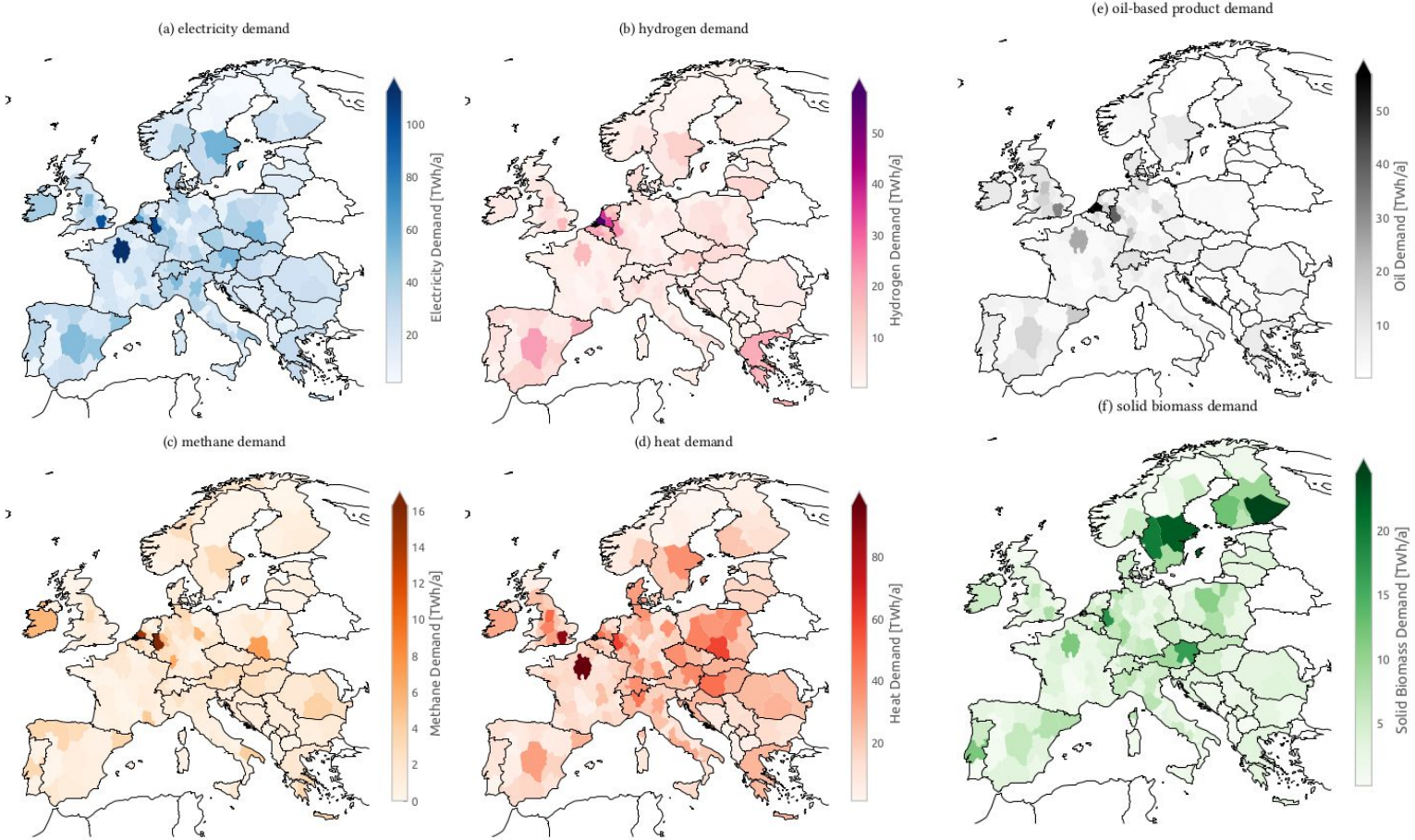
Extension by other sectors requires more data!



Many more rules and data sources!

High-resolution version:
https://cloud.pypsalabs.org/s/ZgkJnns8fsjYcqk

Spatial distribution of energy demands



Use of PyPSA components

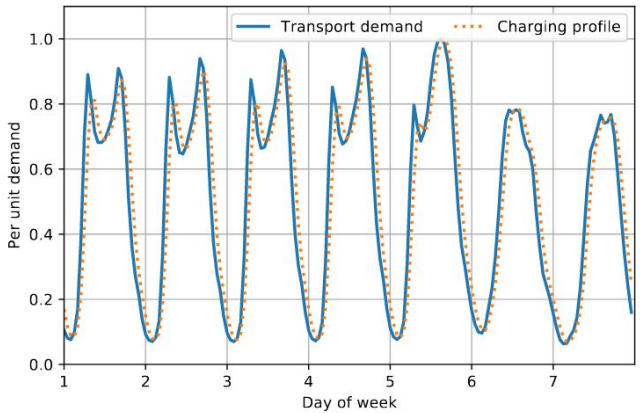
- **Buses** impose energy balances for different energy carriers (heat, hydrogen, methane), and account for CO_2 emitted/absorbed from the atmosphere.

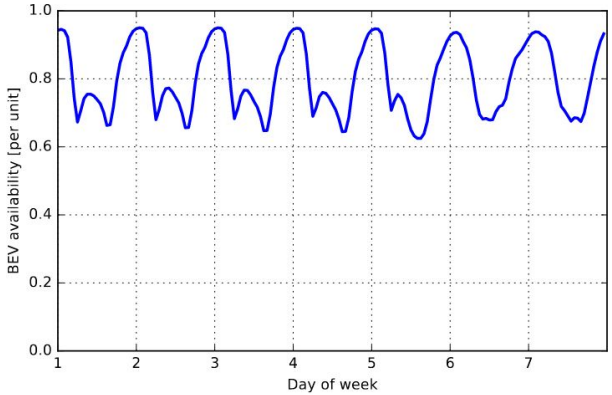
- **Links** represent **energy converters**, e.g. a heat pump that withdraws energy from the electricity bus and supplies heat to the heating bus with a certain efficiency.

- **Multilinks** represent more **elaborated conversion processes**, e.g. a biomass-fired CHP that withdraws biomass and supplies electricity, heat and CO_2 emissions.

- **Stores** can represent available potential (if allowed to discharge throughout the modelled period) or relax the hourly energy balancing constraint to annual (when a free store is added).

Transport - Electrification of land transport

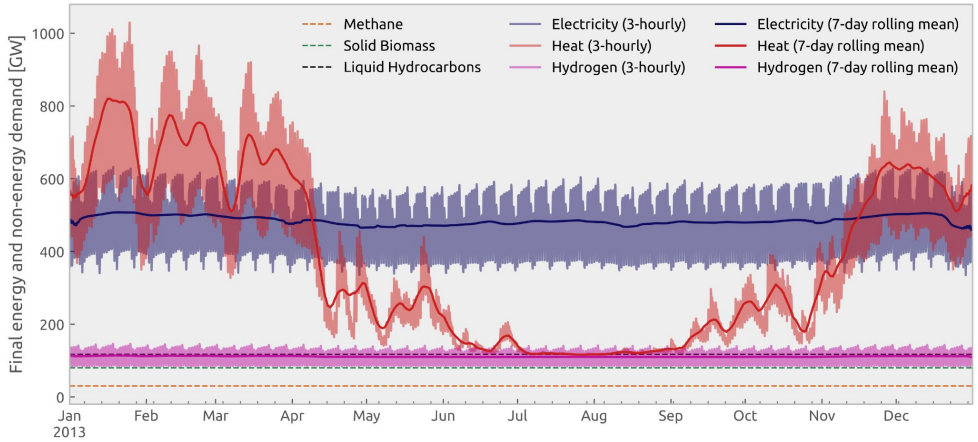




- Road and rail are **largely electrified**, ICE only during pathway to net-zero
- Time series from traffic counting
- Lumped representation of electric vehicles: a configurable share of them participates in demand-side management (**DSM**) or vehicle-to-grid (**V2G**)
- Electric vehicles have **time-varying availability** for charging depending on driving profiles
- EV battery almost fully charged in the morning

Modelling **land transport transformation** endogenously is currently under development!

Heating - Demand



Heating demand time series created by :

- space heating based on degree-day and daily profiles, weighted by population
- constant water heating demand
- scaled to historical annual demand
- split into individual and central systems

  (cooling demand already electrified)

Heating - Tech for individual & district heating

Decentral individual heating
can be supplied by:

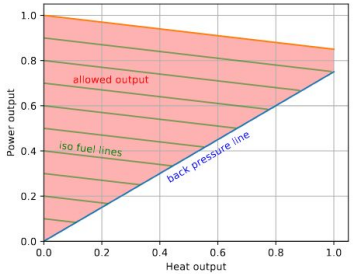
- air- or ground-sourced heat pumps
- resistive heaters
- gas / oil / biomass / hydrogen boilers
- solar thermal
- small water tanks

Building renovations can be co-optimized to reduce space heating demand.

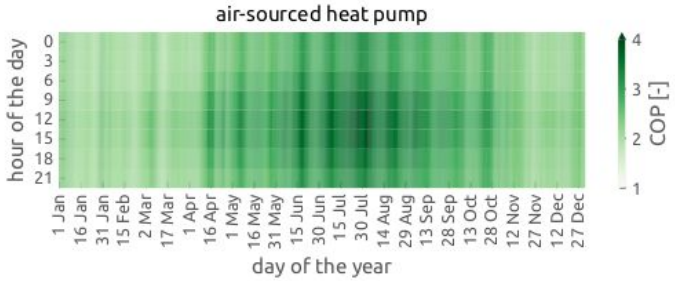
District heating systems
can be supplied in urban areas by:

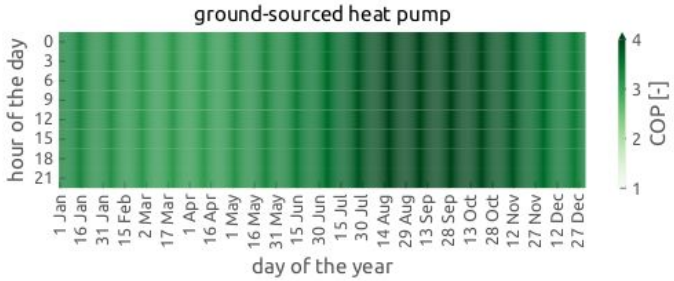
- air-sourced heat pumps + other
- resistive heaters
- gas / hydrogen / biomass / waste CHPs
- gas / oil / biomass / hydrogen boilers
- solar thermal
- long-duration hot water storage
- waste heat from industrial processes

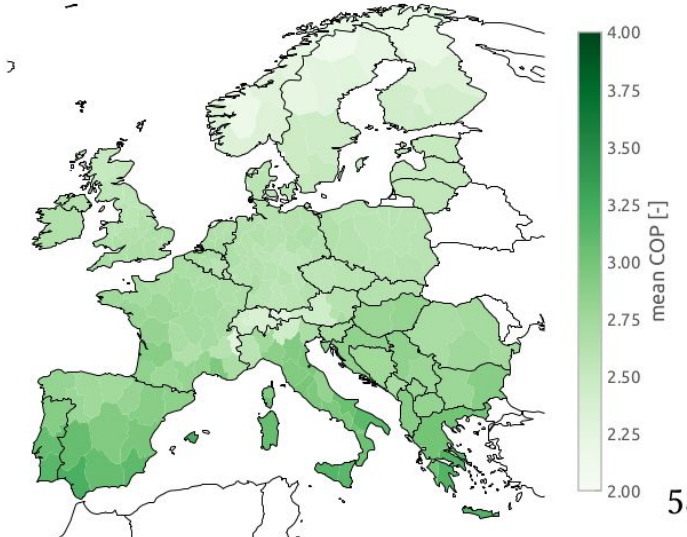
CHP feasible dispatch:

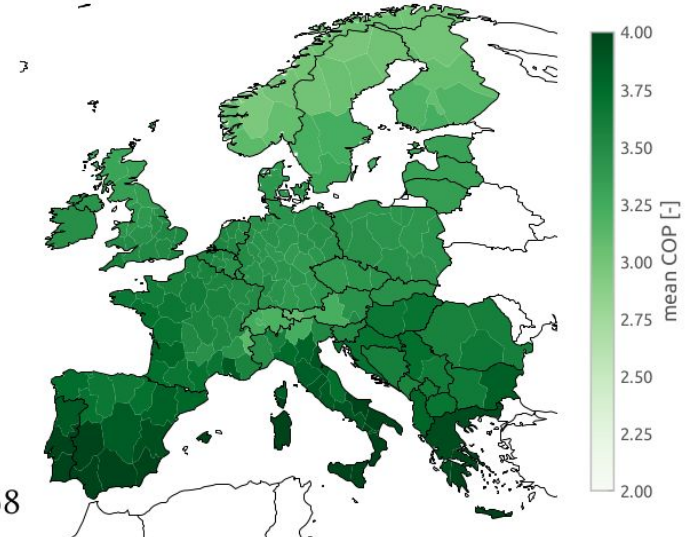


Heating - Heat pumps as new variable supply tech



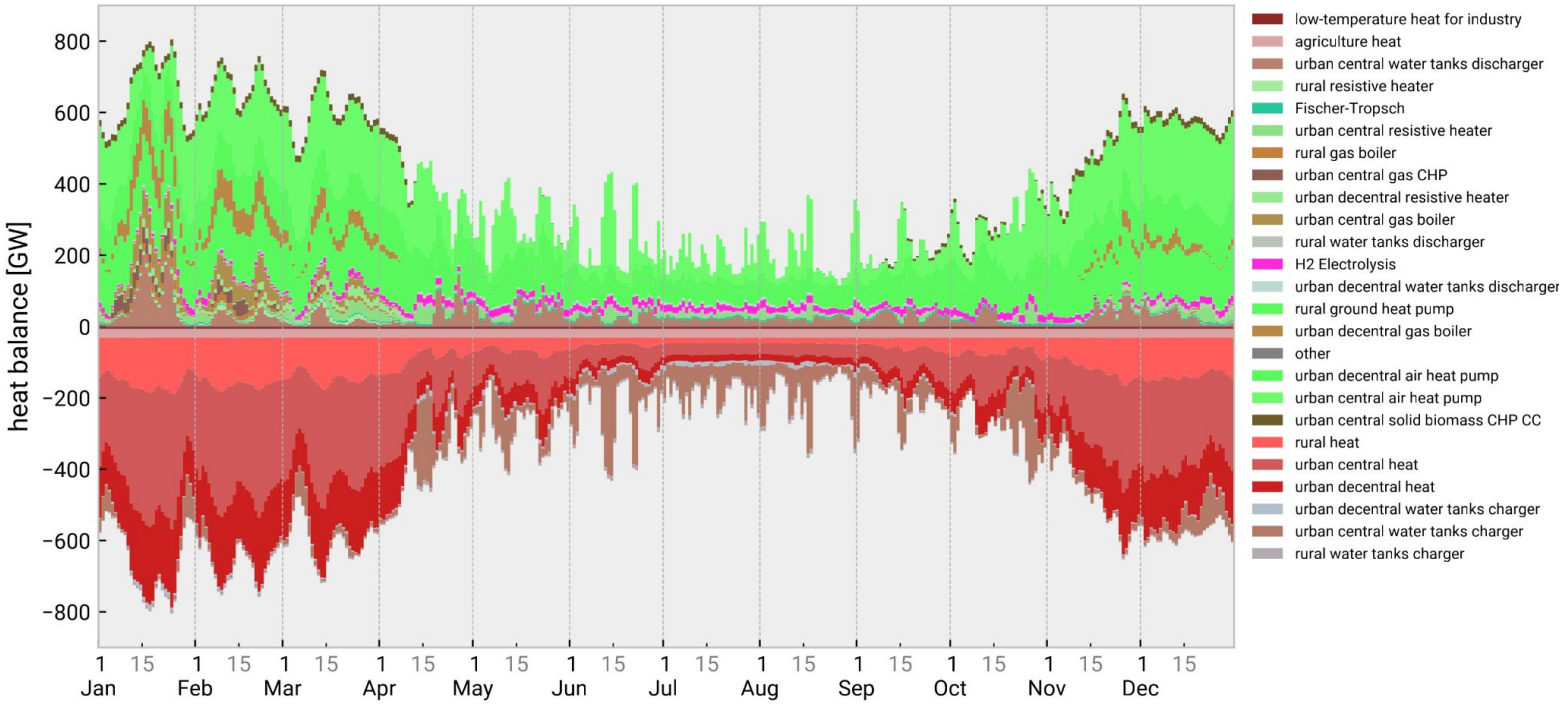






Geothermal heat sources have been integrated very recently!

Heating - Example daily heat system balance



There are difficult periods in winter with low wind and solar, high space heating demand and low air temperatures, which are bad for air-sourced heat pump performance. In this case **gas boilers** and **CHP plants** jump in as backup.

Synthetic fuels - Power-to-X

Carbonaceous fuels produced synthetically:

- Synthetic methane using Sabatier reaction

  $CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$

- Synthetic oil via Fischer-Tropsch

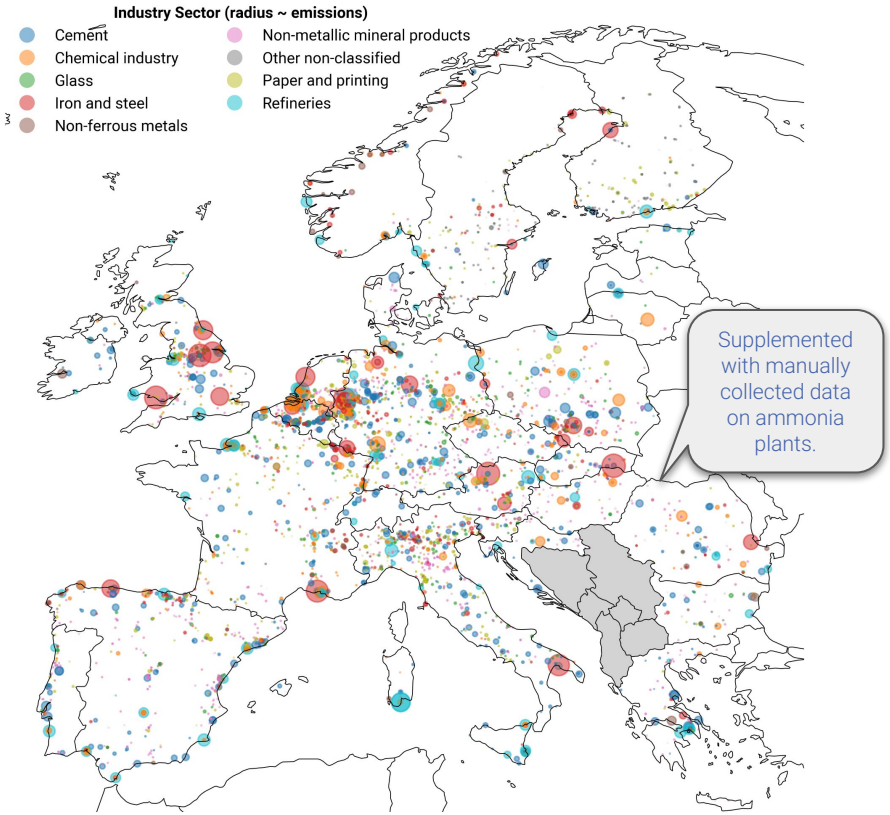
  $(2n+1)\ H_2 + n\ CO \rightarrow C_nH_{2n+2} + n\ H_2O$

- Synthetic methanol in methanolisation plants

  $CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$

Conversion efficiencies read from open database of techno-economic assumptions
`https://github.com/PyPSA/technology-data/blob/master/outputs/costs_2030.csv`

Industry - Regionalisation based on Hotmaps



https://gitlab.com/hotmaps/industrial_sites/industrial_sites_Industrial_Database

Iron & Steel	Phase-out integrated steelworks; increased recycling; rest from H_2-DRI + EAF
Aluminium	Methane for high-enthalpy heat; increased recycling
Cement	Solid biomass; capture of CO_2 emissions
Ceramics	Electrification
Ammonia	Gray, blue, green hydrogen
Plastics	Synthetic naphtha; MtO/MtA, increased recycling
Other industry	Electrification; process heat from biomass
Shipping	Methanol, (oil), (liquid hydrogen), (LNG)
Aviation	Kerosene from Fischer-Tropsch or methanol

Modelling **industry relocation, high-temperature heat source & shipping fuels** endogenously is currently under development!

Infrastructure - Gas network with H_2 retrofitting

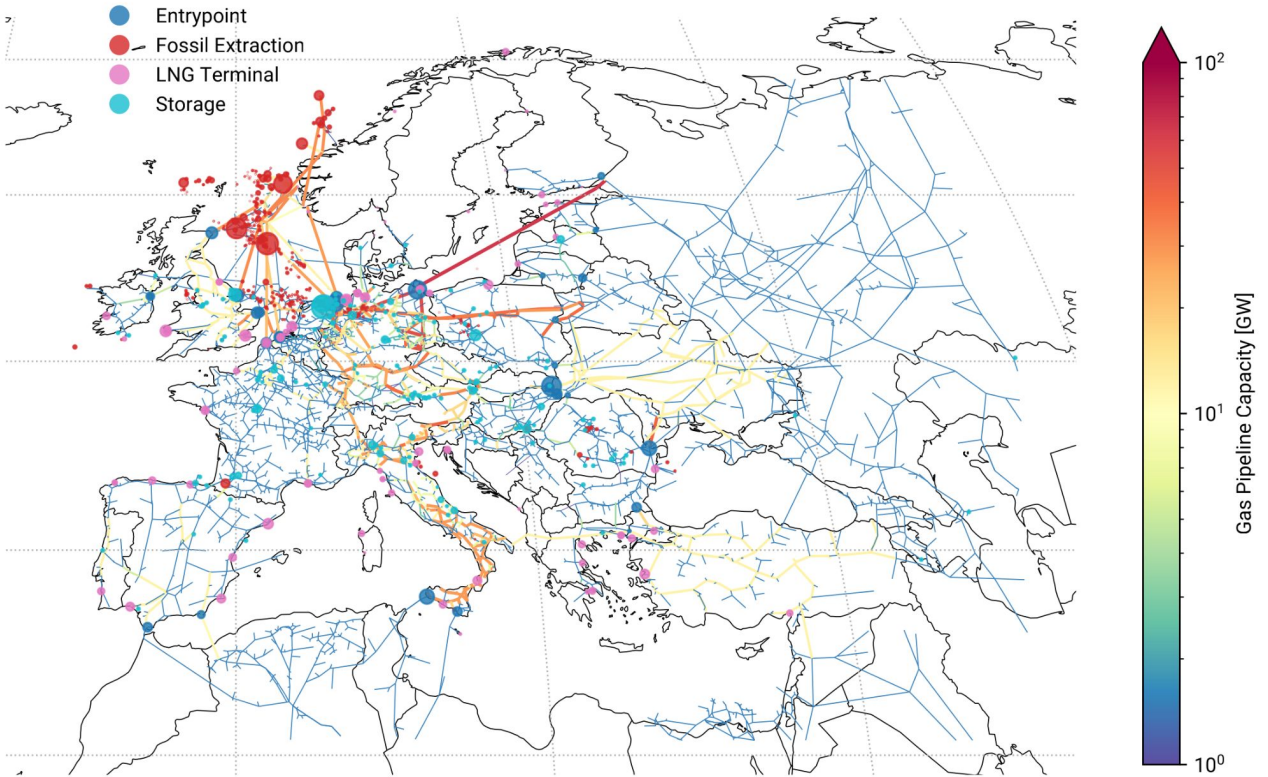
Compiled from open **SciGRID_gas** dataset.

Fossil gas enters at **LNG terminals** or **gas fields**.

Gas flow **physics** and **valve control** neglected transport model.

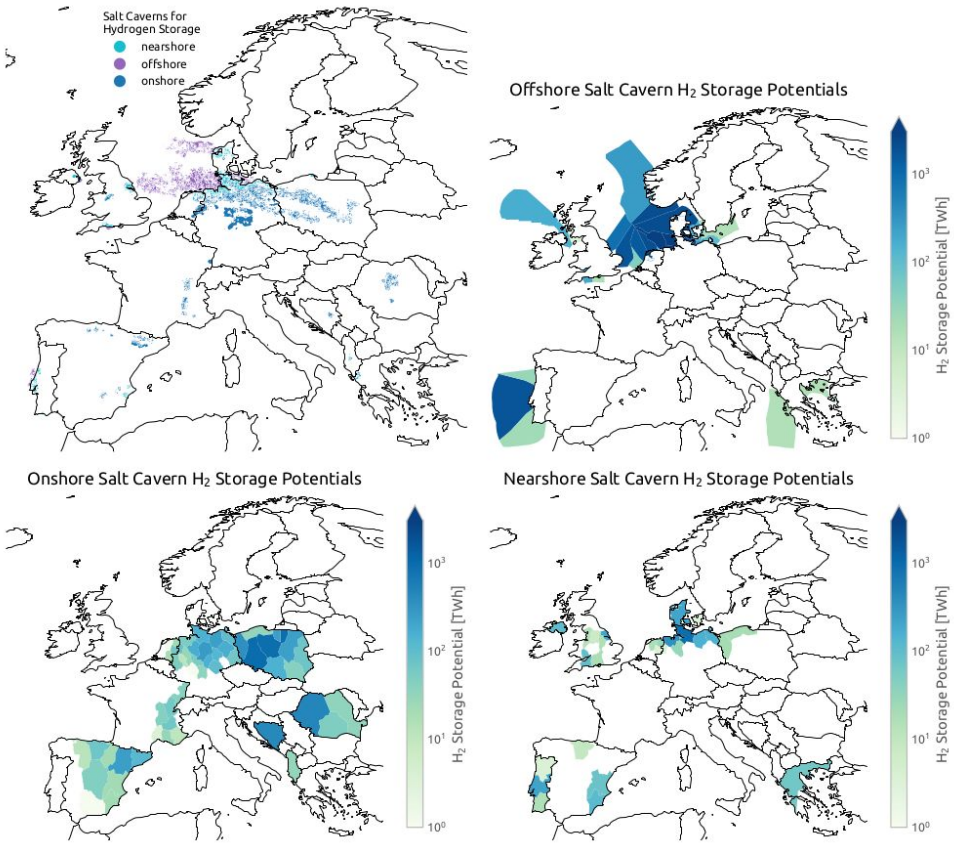
Electricity demand for **compression** and **leakage** configurable.

Pipelines can be **retrofitted** to H_2 with costs from EHB.



https://zenodo.org/records/4767098

Infrastructure - Hydrogen storage potentials

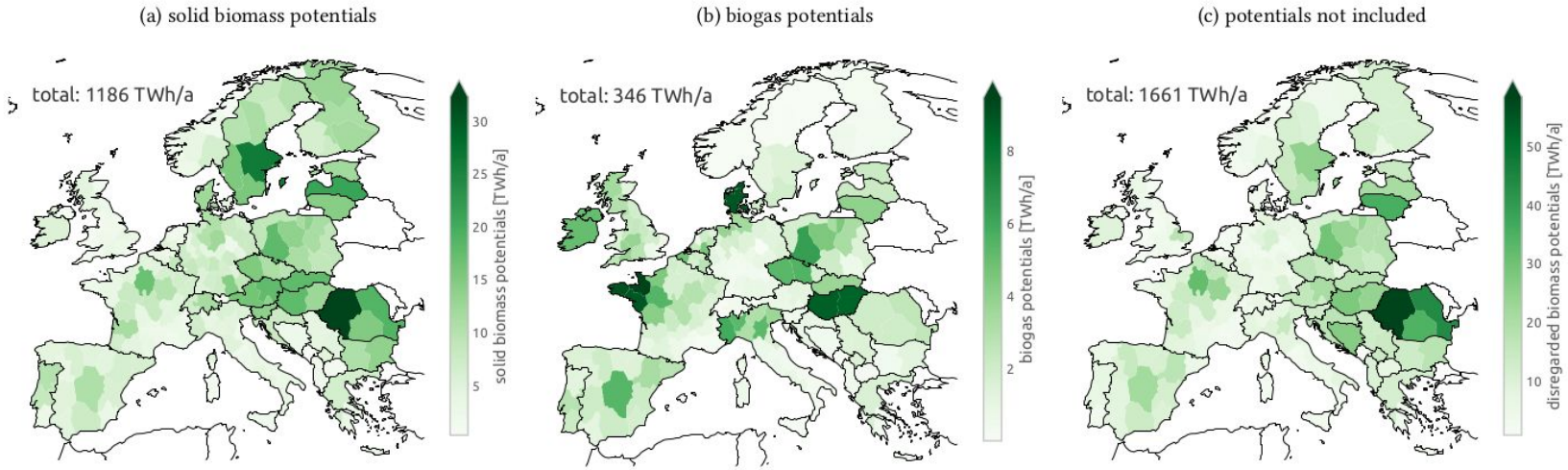


The regional distribution of **geological potential** to store hydrogen in **salt caverns** is considered.

The user can **configure** if onshore and/or offshore potentials can be used.

Dilara Gulcin Caglayan, Nikolaus Weber, Heidi U. Heinrichs, Jochen Linßen, Martin Robinius, Peter A. Kukla, Detlef Stolten, *Technical potential of salt caverns for hydrogen storage in Europe*, **International Journal of Hydrogen Energy**, Volume 45, Issue 11, 2020, 6793-6805, https://doi.org/10.1016/j.ijhydene.2019.12.161

Infrastructure - Biomass from JRC ENSPRESO



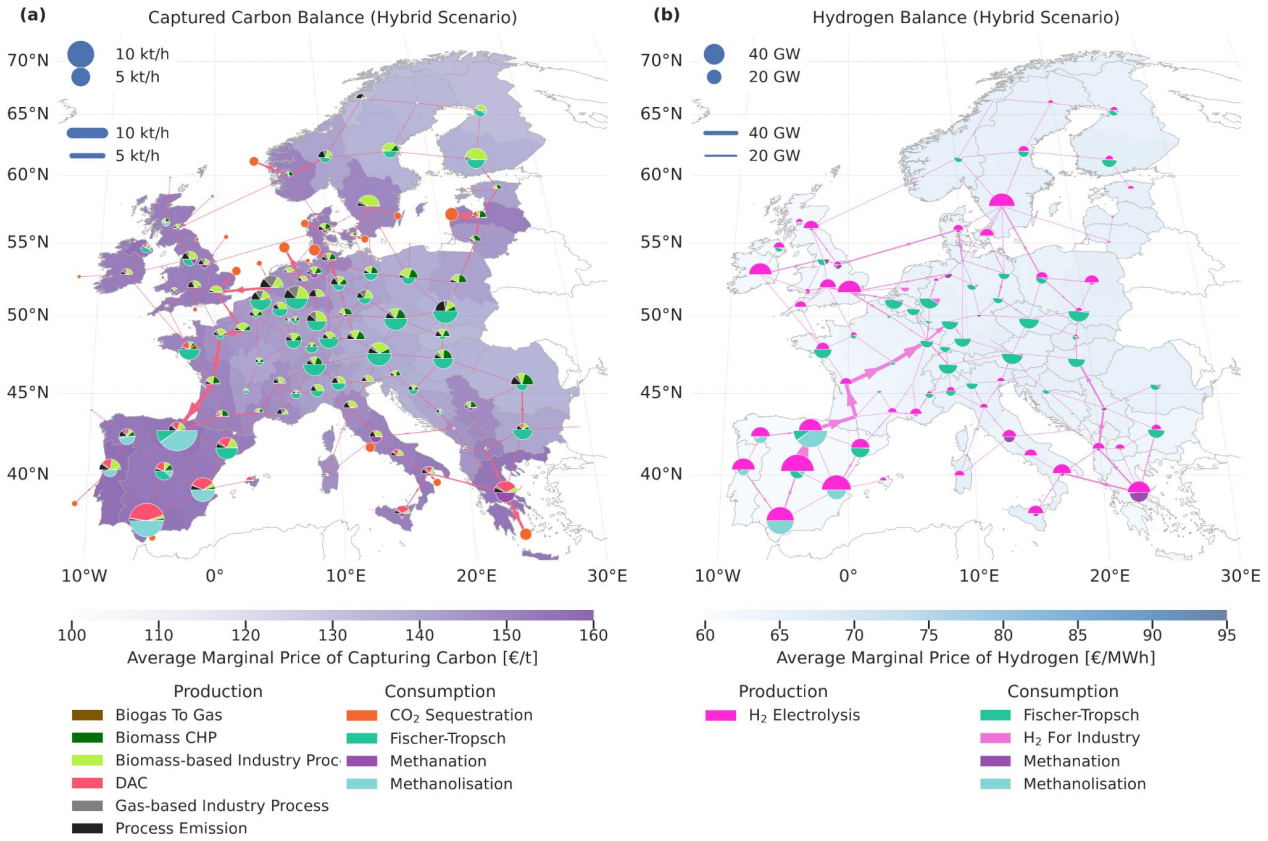
Biomass potentials are split between **solid biomass** and **biogas** (which can be, for instance, upgraded).

The user can configure low/medium/high potentials and what categories of biomass to consider (e.g. forest residues).

The default configuration only considers **residual biomass**, no energy crops.

https://data.jrc.ec.europa.eu/dataset/74ed5a04-7d74-4807-9eab-b94774309d9f

Infrastructure - Carbon management

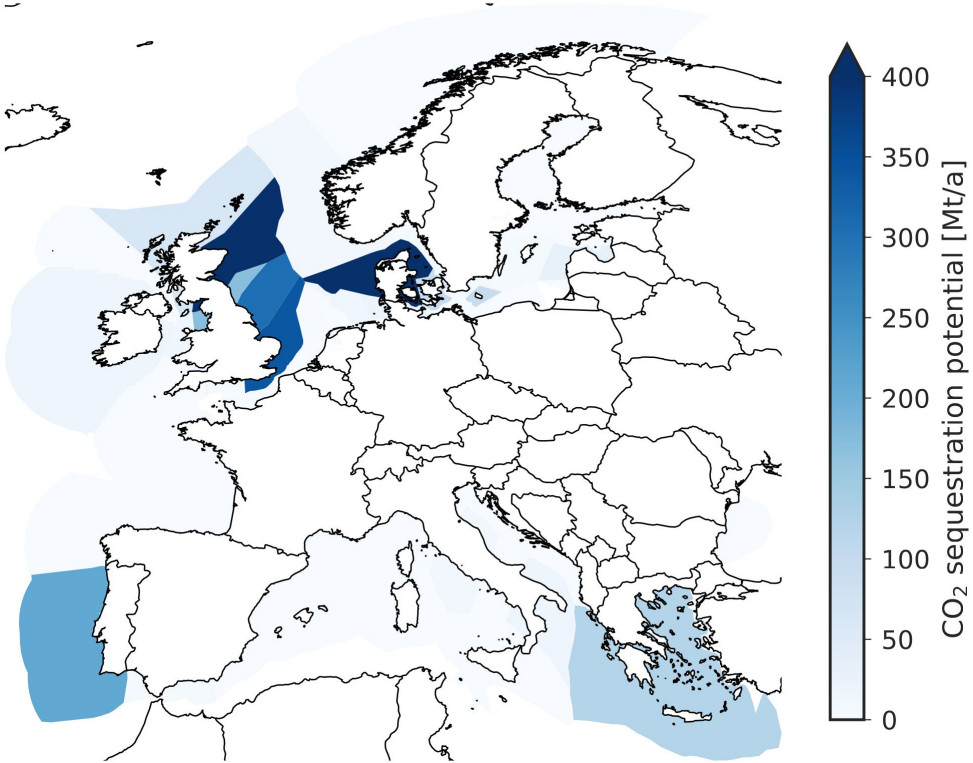


Built-in carbon flows:

- **Capture:** DAC, process emissions, fossil / biomass CHP
- **Transport:** CO_2 pipelines
- **Storage:** intermediate storage and long-term geological sequestration
- **Utilization:** for synthetic carbonaceous fuels

Infrastructure - Carbon sequestration potentials

Example: Offshore carbon sequestration potentials



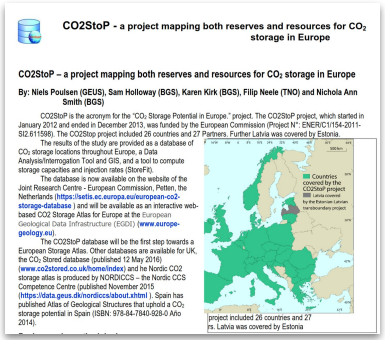
The user can **configure**

- onshore/offshore sequestration,
- gas fields/oil fields/aquifer, and
- low/medium/high potentials,

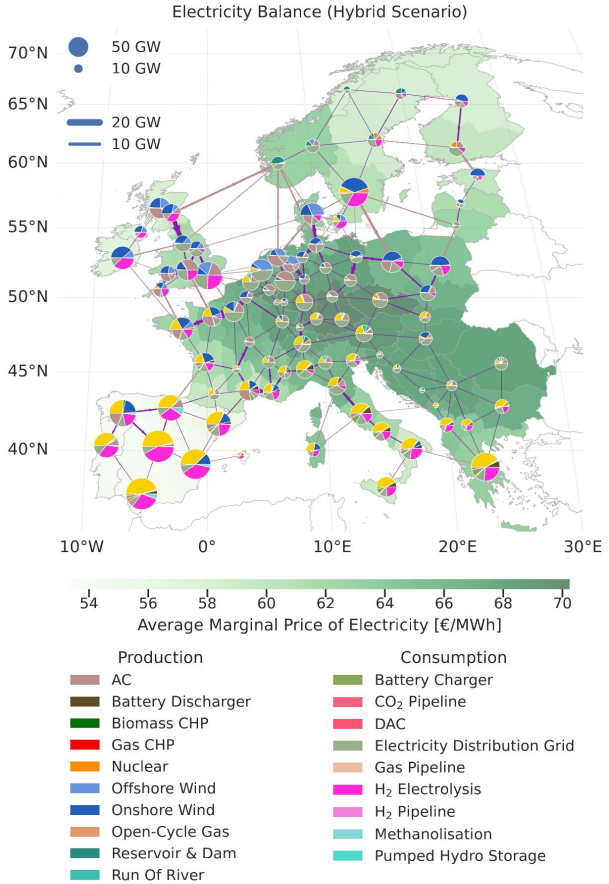
as well as a **total limit** on the annual sequestration, e.g. 250 Mt per year.

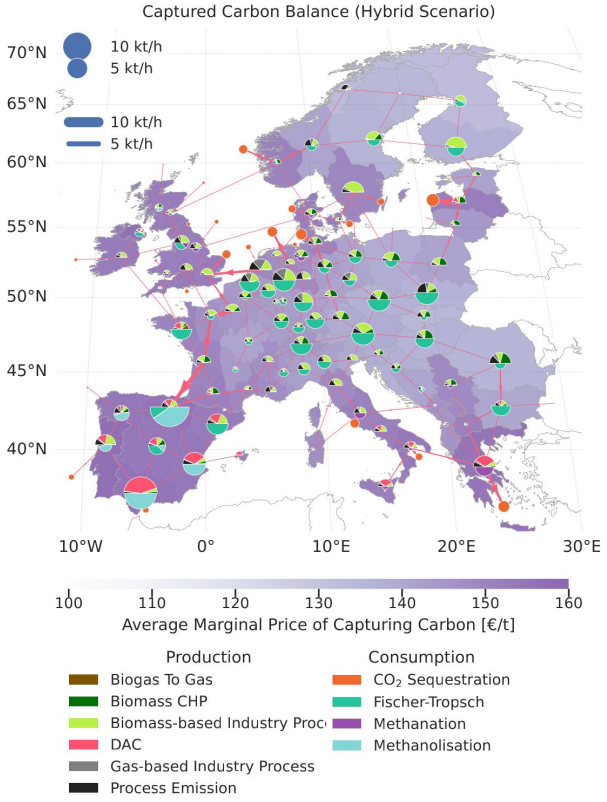
Data source:

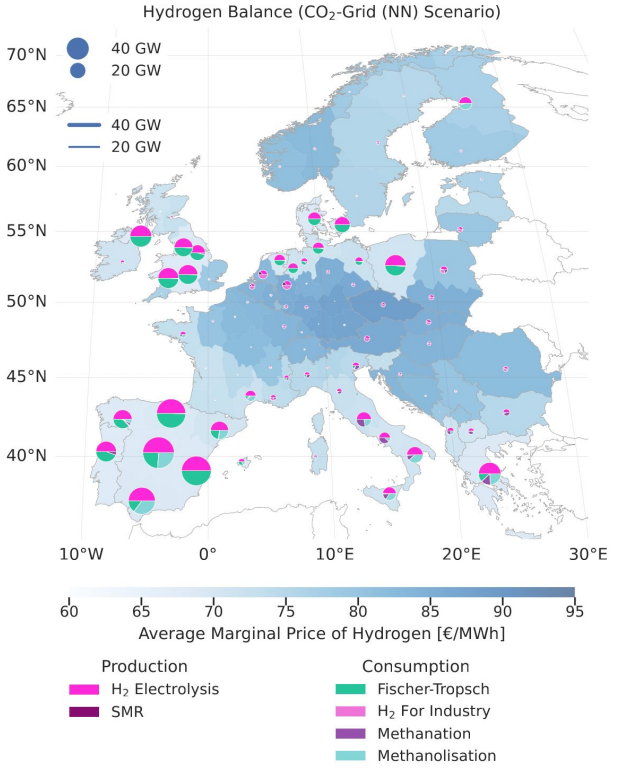
https://energy.ec.europa.eu/publications/assessment-co2-storage-potential-europe-co2stop_en



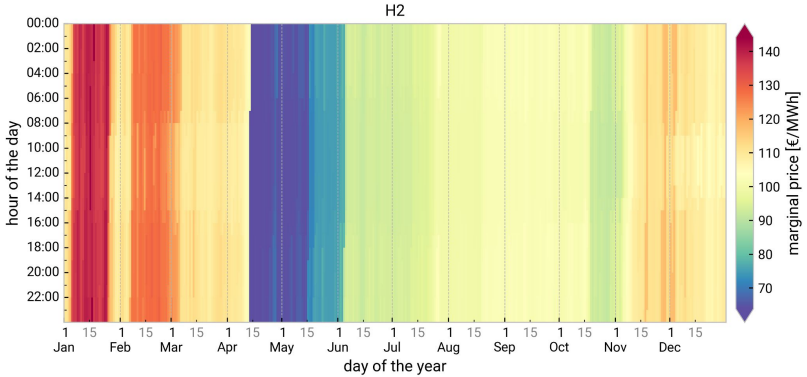
Examples of sector-coupled results

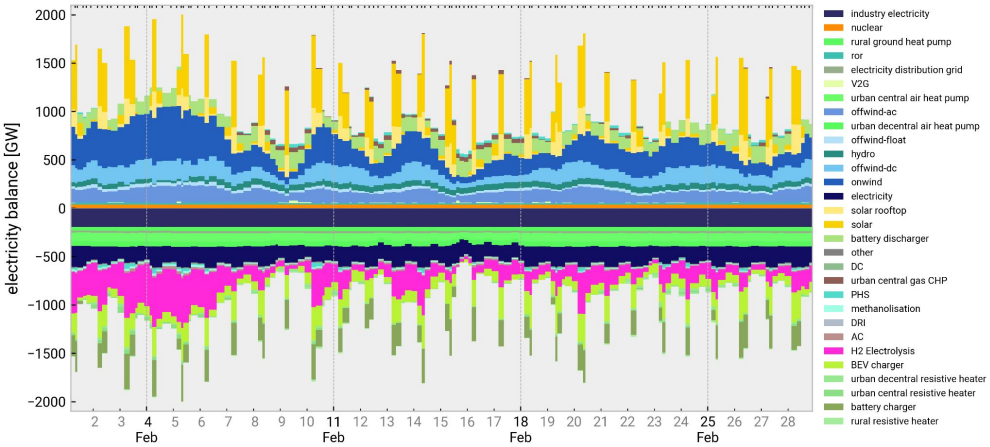


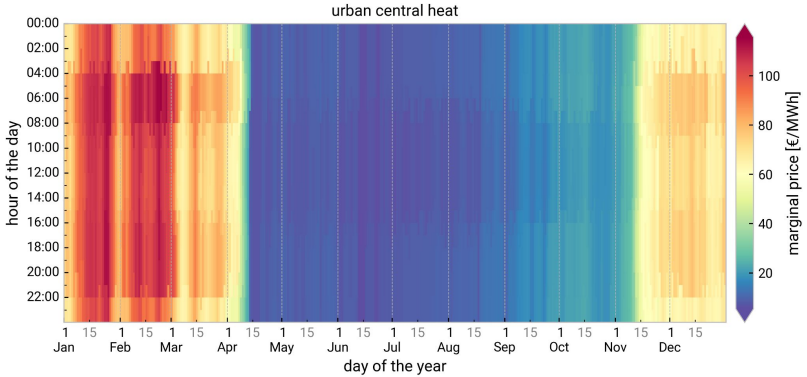


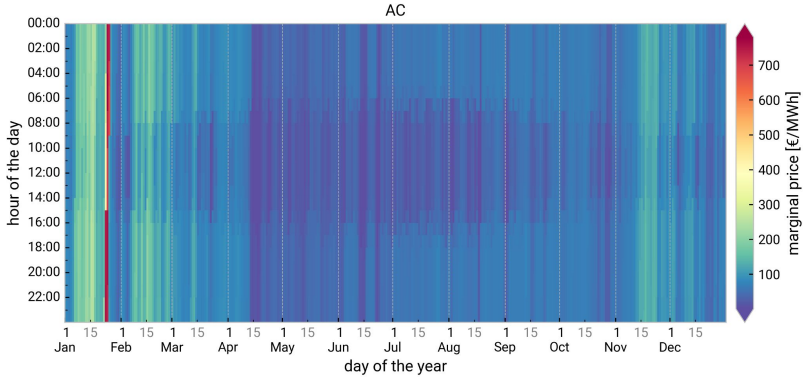


Prices in the model









Technology choices - endogenous vs. exogenous

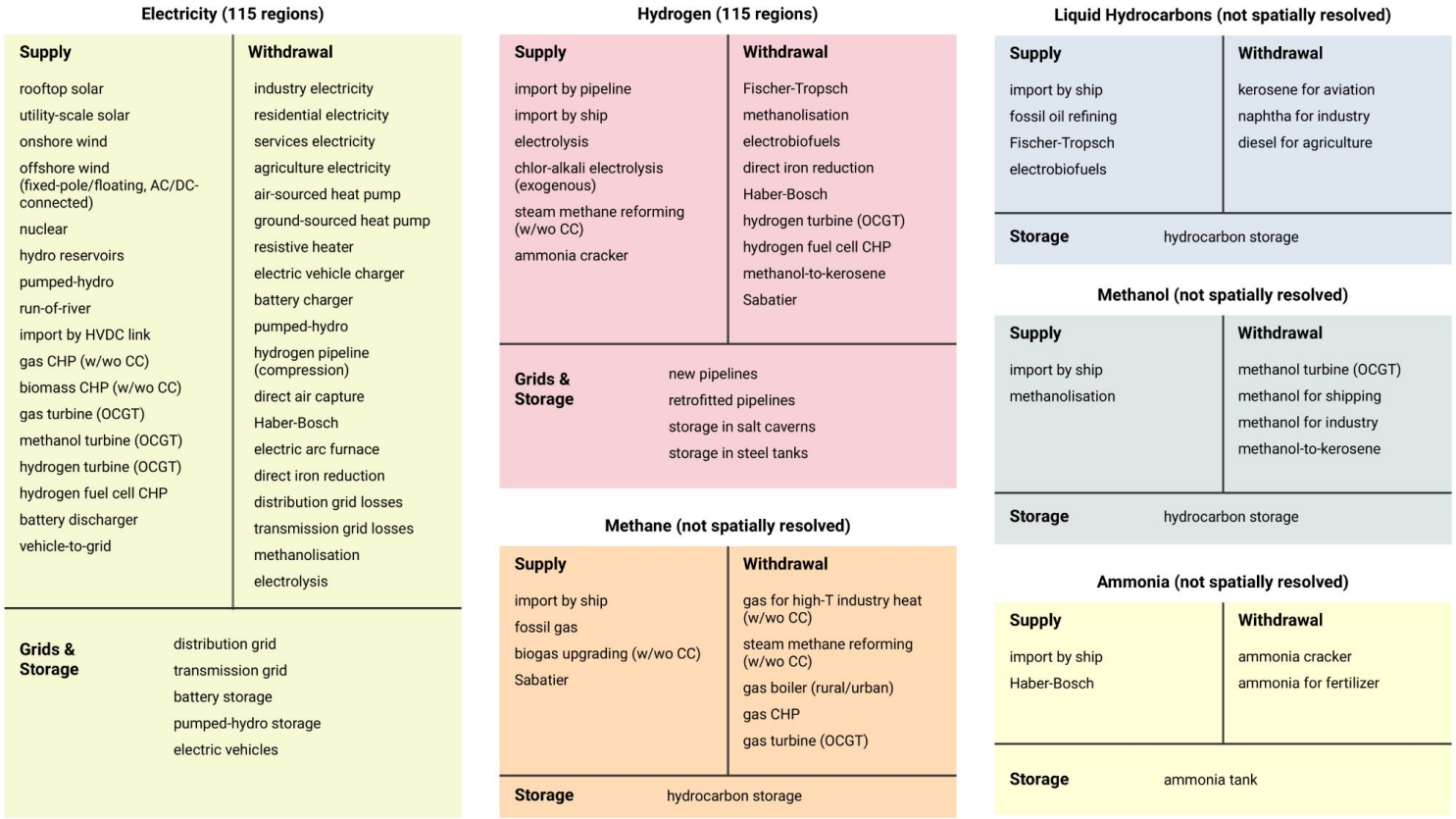
Exogenous assumptions (modeller chooses):

- energy services demand (e.g. heat)
- district heating shares
- energy carrier shares for road transport
- kerosene for aviation
- methanol for shipping
- electrification & recycling in industry
- steel production with DRI + EAF

Endogenous choices (model optimizes):

- change in electricity generation fleet
- transmission reinforcement
- capacities and locations of short and long-duration energy storage
- space and water heating technologies (including building renovations)
- all P2G/L/H/C
- supply of process heat for industry
- carbon capture (e.g. CHP, industry, DAC)

Supply, consumption and storage options by carrier



Supply, consumption and storage options by carrier

Heat (115 regions)

Supply	Withdrawal
air-sourced heat pump	residential heat
ground-sourced heat pump (only rural)	services heat
resistive heater	agriculture heat
gas boiler	low-T industry heat
biomass boiler	direct air capture
solar thermal	water tank charger
water tank discharger	
biomass CHP (w/wo CC, only DH)	
gas CHP (w/wo CC, only DH)	
hydrogen fuel cell CHP (only DH)	
electrolysis (only DH)	
Haber-Bosch (only DH)	
Sabatier (only DH)	
Fischer-Tropsch (only DH)	
methanolisation (only DH)	

Storage	
	long-duration thermal storage (only DH)
	hot water tank

CO2 atmosphere (not spatially resolved)

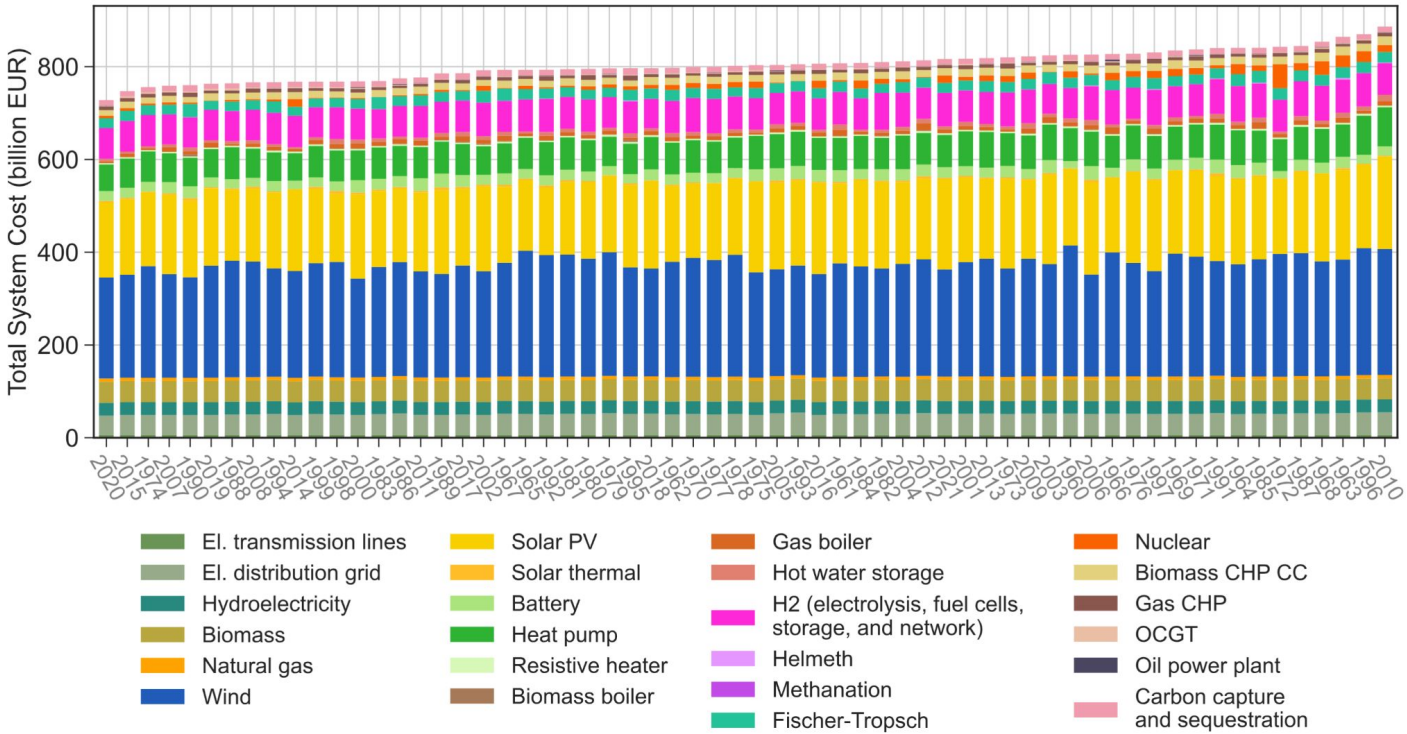
Supply	Withdrawal
kerosene for aviation	solid biomass for industry (w CC)
diesel for agriculture	solid biomass CHP (w CC)
methanol for shipping	biogas upgrading (w CC)
methanol for industry	direct air capture
naphtha for industry	electrobiofuels
gas boiler	
gas CHP (w/wo CC)	
gas turbine (OCGT)	
methanol turbine (OCGT)	
process emissions (w/wo CC)	
fossil oil refining	
gas for high-T industry heat (w/wo CC)	
steam methane reforming (w/wo CC)	

CO2 commodity (not spatially resolved)

Supply	Withdrawal
direct air capture	Fischer-Tropsch
biogas upgrading (w CC)	methanolisation
gas CHP (w CC)	sequestration
biomass CHP (w CC)	Sabatier
steam methane reforming (w CC)	
process emissions (w CC)	
solid biomass for industry (w CC)	
gas for high-T industry heat (w CC)	

Storage	
	intermediate storage in steel tank
	long-term geological sequestration

PyPSA-Eur can be run on different **weather years**!



The years **2010**, **2013**, **2019** and **2023** are currently available "out of the box".

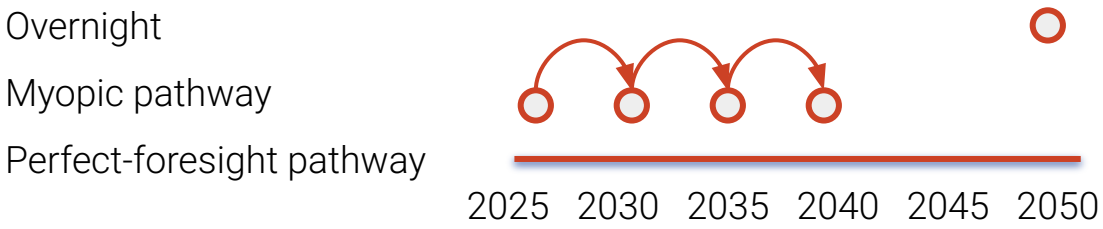
Other years **1940-2024** require a few more steps.

We are planning to expand the number of "plug-and-play" years.

https://www.nature.com/articles/s41467-024-54853-3

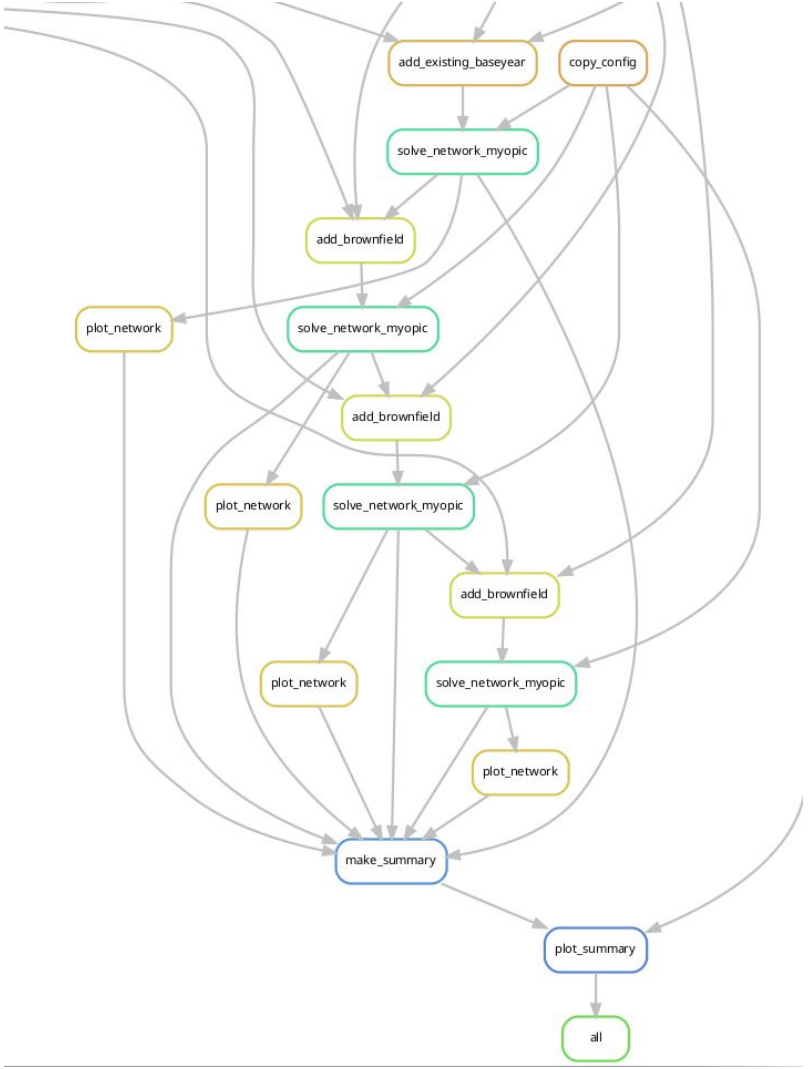
Overnight and pathway optimization

- **Overnight** optimization builds the complete system in one go. Already existing capacities can be included (**brownfield**) or the system can be built from scratch (**greenfield**)
- The choice of **investment years** is arbitrary.
- **Pathway optimization with myopic foresight** (next slide). The CO_2 emission **reduction path** is exogenous.
- **Perfect foresight pathway planning** is currently experimental (i.e. use the CO_2 budget endogenously).



Myopic pathway optimization

- The CO_2 emission **reduction path** is exogenous.
- Optimise **start network** for e.g. 2025, starting with existing energy infrastructure.
- Take results from **2025 as input** for 2030 infrastructure optimisation, take 2030 results for next iteration, etc.
- Infrastructure decommissioned when reaching end of lifetime
- Configurable social discount rate and number of planning horizons



Scenario management

PyPSA-Eur has integrated & scalable scenario management.

config/config.demo..yaml

```yaml
run:
  name: all
  scenarios:
    enable: true

scenario:
  clusters: [90]

sector:
  H2_network: true
  gas_network: true
  H2_retrofit: true

electricity:
  transmission_limit: vopt
```

With these two files configured, run:

```
$ pixi run snakemake --configfile config/config.demo.yaml
```

config/scenarios.yaml

```yaml
no-h2-network:
  sector:
    H2_network: false

no-grid-expansion:
  electricity:
    transmission_limit: v1.0

no-to-both:
  sector:
    H2_network: false
  electricity:
    transmission_limit: v1.0

yes-to-both:
  sector:
    H2_network: true
  electricity:
    transmission_limit: vopt
```

Practical Phase

(sector-coupled)

1) Run PyPSA-Eur sector-coupling tutorial with `snakemake`

   Guide:
   https://pypsa-eur.readthedocs.io/en/latest/tutorial_sector/

   ```
   pixi run snakemake -call all --configfile config/test/config.overnight.yaml
   ```

2) Explore CSV files and images in `results` directory.

Go to https://pypsa-eur.readthedocs.io/en/latest/configuration.html and try to find out how to configure some of the settings for **sector-coupled models** listed below:

1. Disable vehicle-to-grid discharging.
2. Disable hydrogen network expansion.
3. Increase the carbon sequestration potential to 800 Mt/a.
4. Allow hydrogen underground storage also onshore.
5. Reduce the primary production of plastics by increasing recycling rates.

Additional resources

- Materials from the “Integrated Energy Grids” course that we teach at DTU: https://martavp.github.io/integrated-energy-grids/intro.html

- What to expect in the near future? PyPSA roadmap https://pypsa.org/roadmap

(We did not have time to cover it here, but see the nice examples by Fabian and Iegor on)

- Security of Supply Analysis https://pypsalabs.github.io/workshop-202610/berlin/pypsa-scenarios/

- Stochastic & Risk-Averse Optimization https://pypsalabs.github.io/workshop-202610/berlin/pypsa-stochastic/